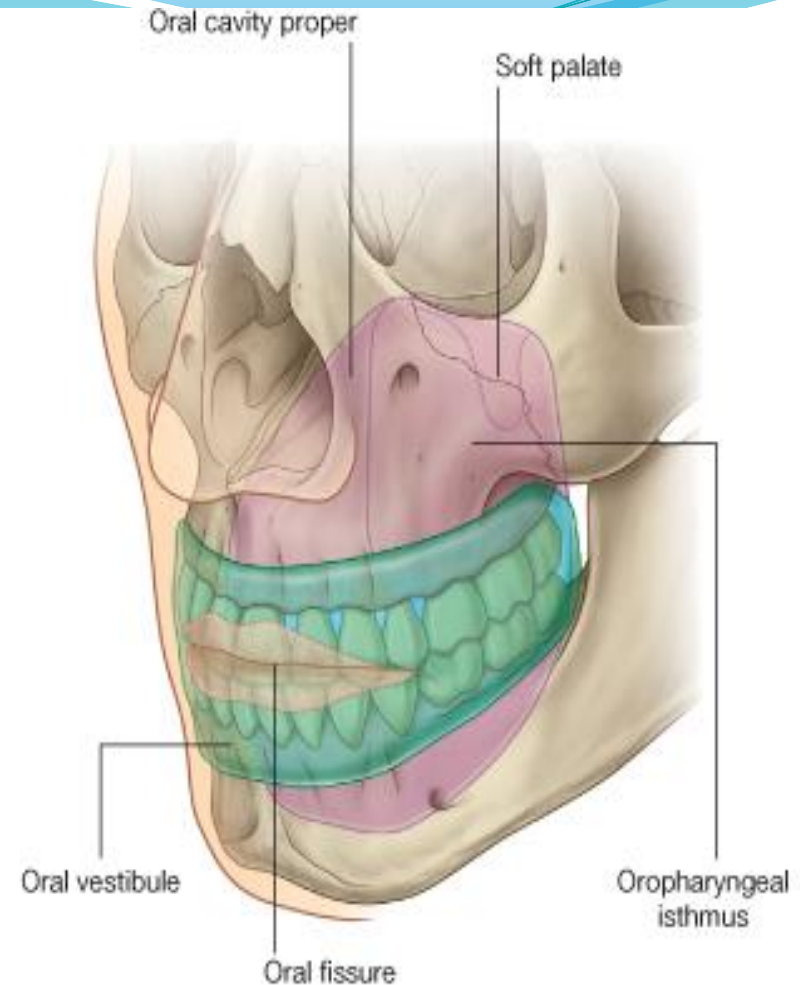




THE ORAL REGION

ORAL REGION

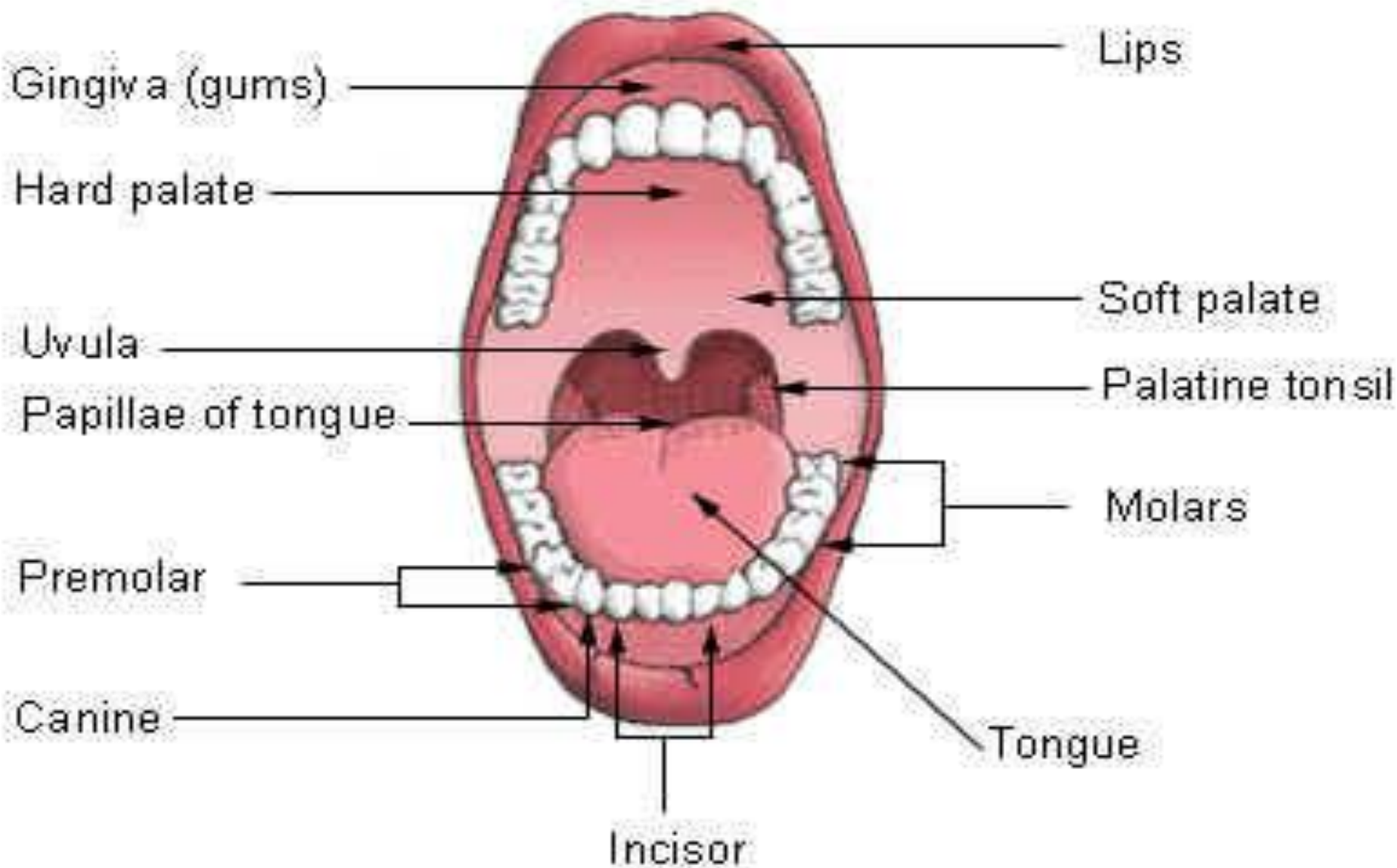
- includes the oral cavity, teeth, gingivae, tongue, palate, and the region of the palatine tonsils
- **The oral cavity:**
 - is the entry portal of the digestive system
 - is bounded anteriorly by the lips and posteriorly by the oropharyngeal isthmus (isthmus faucium), a more or less circular aperture that guards the entrance to the pharynx



• The oral cavity has multiple functions:

- it is the inlet for the digestive system involved with the initial processing of food
- is aided by secretions from salivary glands
- it manipulates sounds produced by the larynx and one outcome of this is speech
- can be used for breathing because it opens into the pharynx, which is a common pathway for food and air
- *For this reason, the oral cavity can be used by physicians to access the lower airway*
- *and dentists use 'rubber dams' to prevent debris such as tooth fragments from passing through the oropharyngeal isthmus and pharynx into either the esophagus or the lower airway*

Mouth (Oral Cavity)



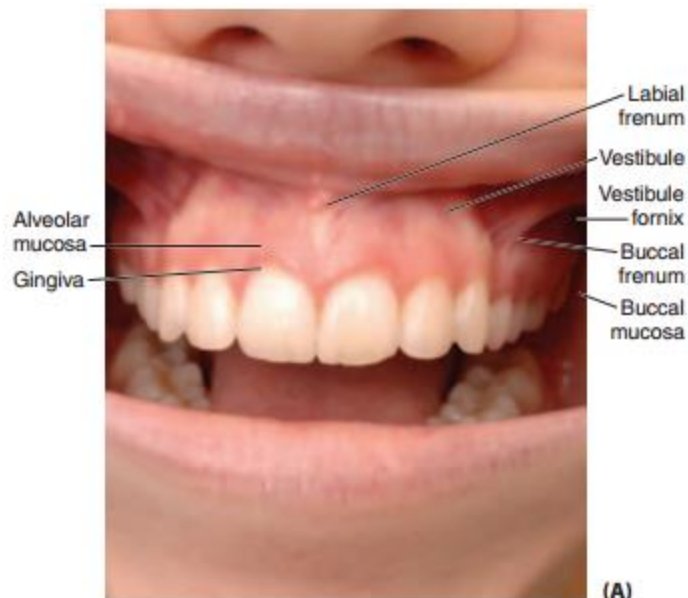
Oral Cavity

- **consists of two parts:**

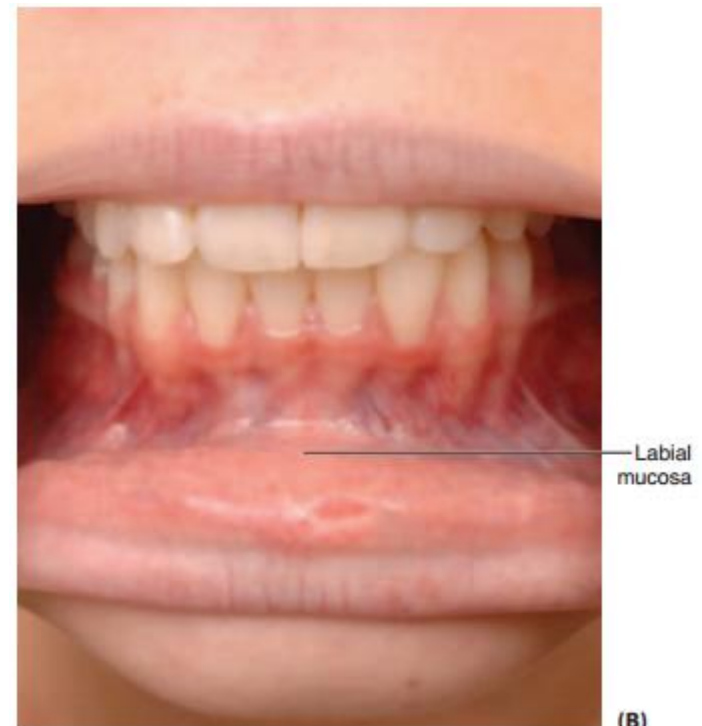
- the oral vestibule and the oral cavity proper

- a. **The oral vestibule**

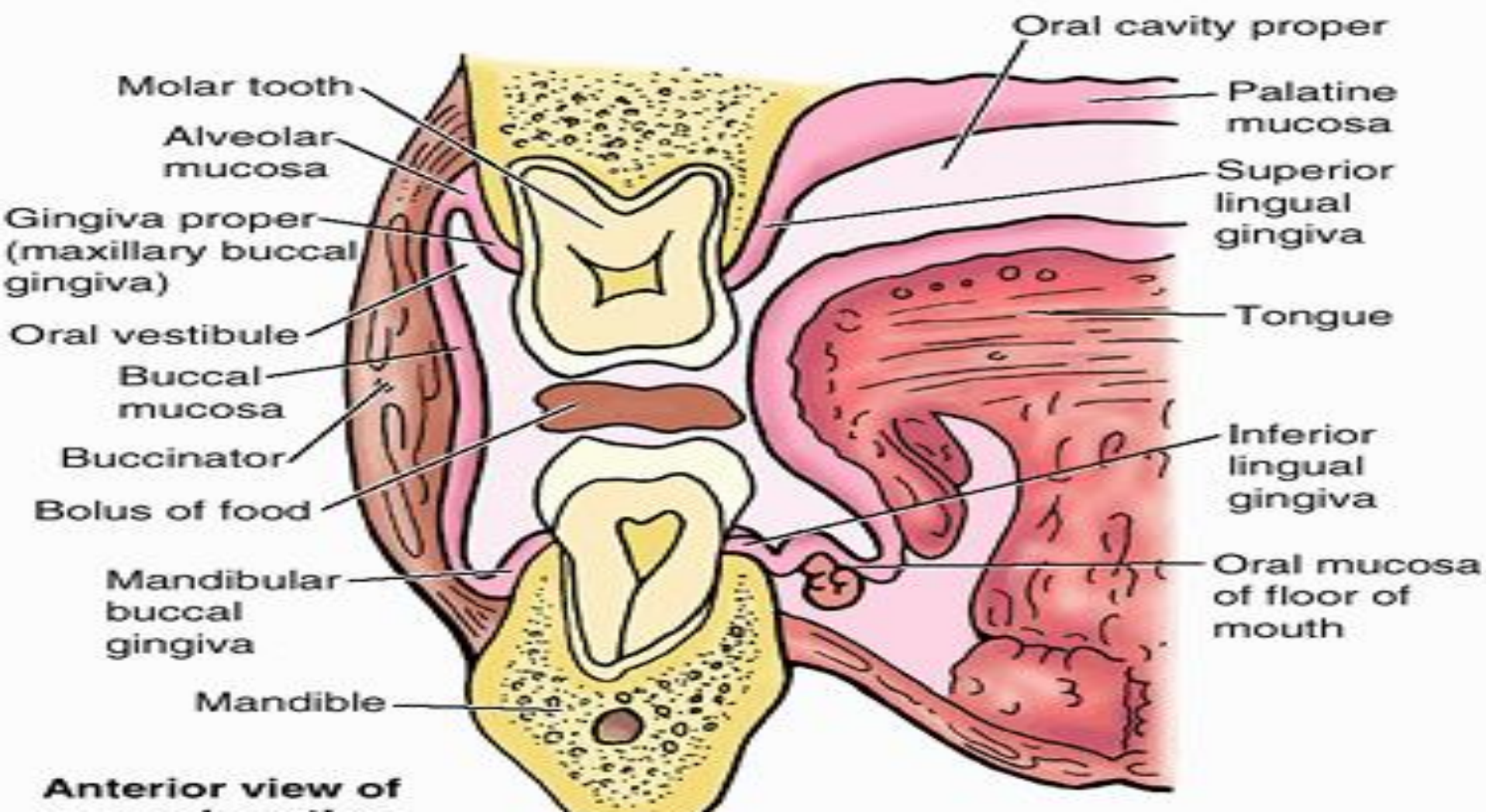
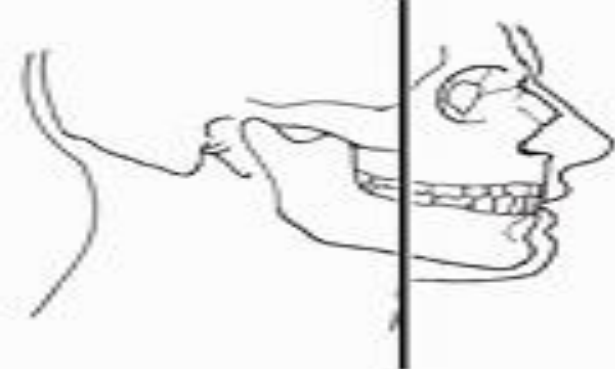
- is the slit-like space between the teeth and gingivae (gums) and the lips and cheeks.
- The vestibule communicates with the exterior through **the oral fissure** (opening).
- The size of the oral fissure is controlled by the circumoral muscles, such as the **orbicularis oris** (the sphincter of the oral fissure), the **buccinator, risorius**, and **depressors and elevators of the lips** (dilators of the fissure).



(A)



(B)

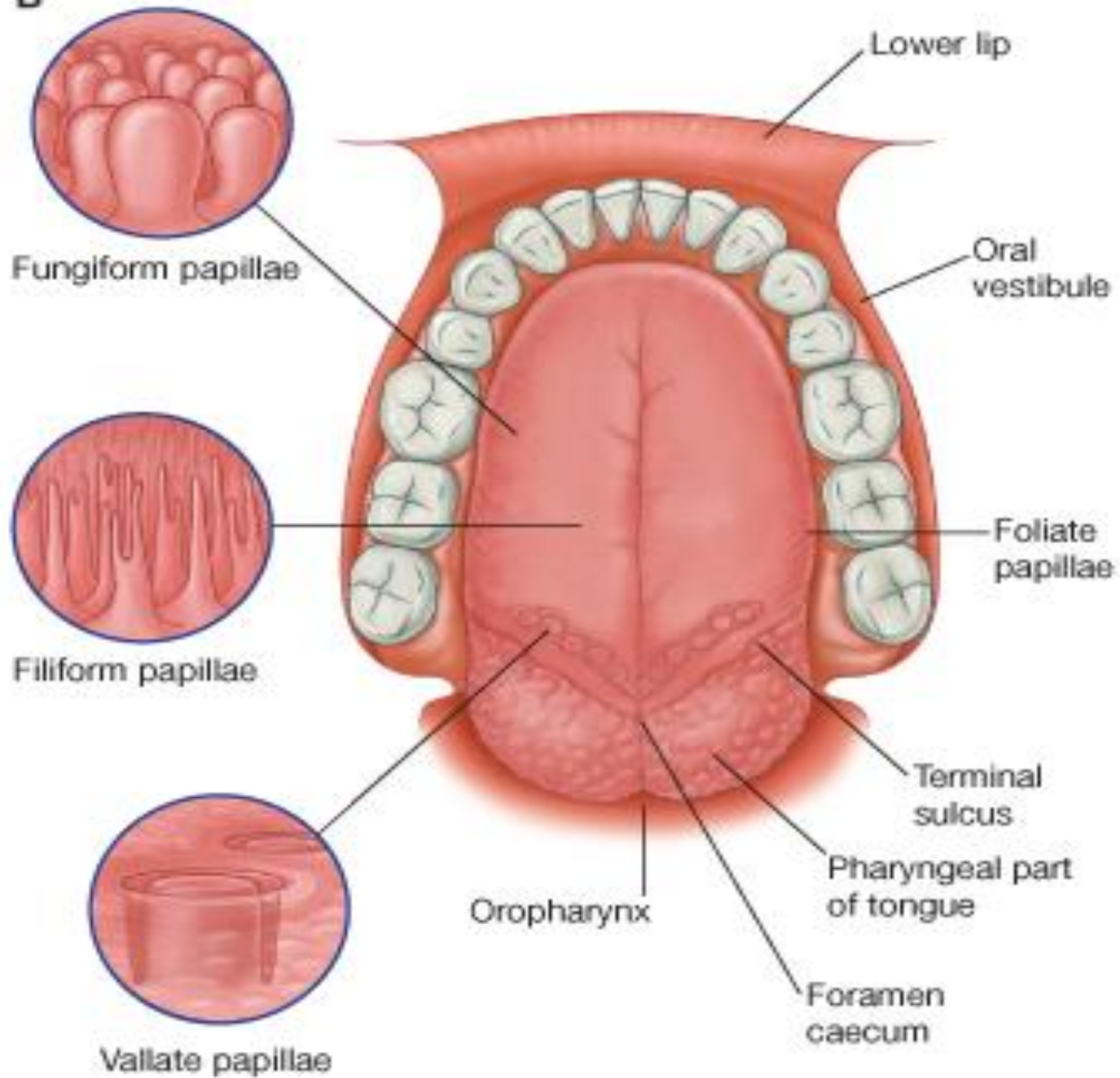


VESTIBULE

- is the cleft or space between the lips and cheeks externally
- and the teeth and gingiva of the dental arches internally
- The vestibule communicates with the exterior through the **oral fissure of the lips**
- with the oral cavity proper via the **interdental spaces** the interval posterior to the last molar teeth in each dental arch
- Laterally, the vestibule is referred to as the **buccal vestibule**,
- Located in the superior labial vestibule is **the incisive fossa** represented by a shallow **depression superior to the incisor teeth**.

- The two lips are connected laterally by the **labial commissures**
- Occasionally a slight depression is noted in the center of the labial commissure, known as the **commissural lip pit**
- The oral fissure (**rima of the mouth**) is the zone between the superior and inferior lips

B



Lips, Cheeks, and Gingivae

- Lips and Cheeks
 - The lips are mobile, musculofibrous folds surrounding the mouth, extending from the **nasolabial sulci** and **nares** laterally
 - and superiorly to the **mentolabial sulcus** inferiorly
- The **transitional zone** of the lips
- **Arterial supply:**
 - The upper lip is supplied by:
 - superior labial branches of **the facial** and **infra-orbital arteries**.
 - The lower lip is supplied by:
 - inferior labial branches of the facial and mental arteries



Vestibular
(mucolabial) fold

Labial frenulum

Labial mucosa

Alveolar mucosa

Superior labial
gingiva

Vermilion
border of lip

(A)

Vestibular
(mucolabial) fold

Labial mucosa

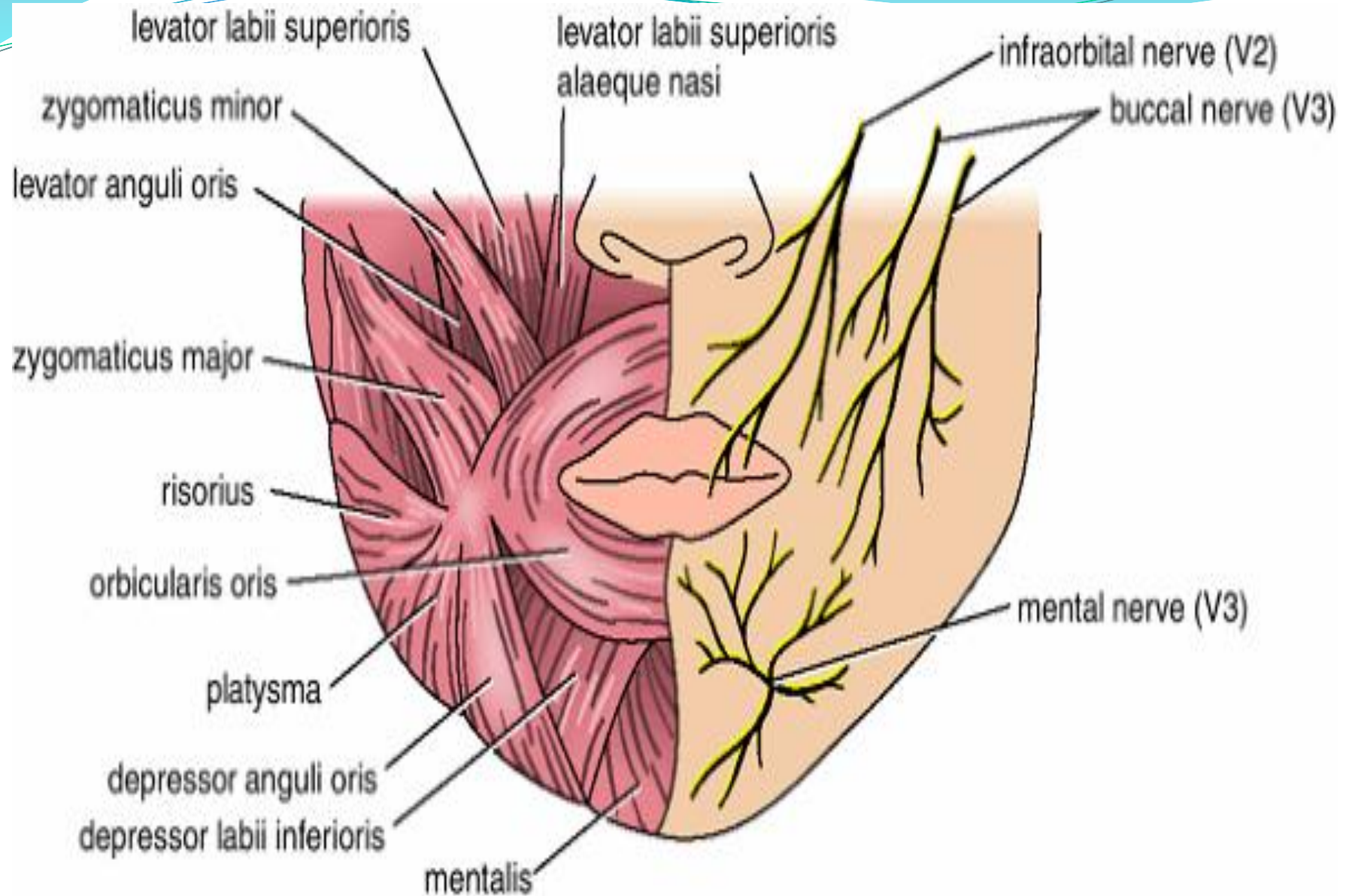
Inferior labial
gingiva (gingiva
proper)

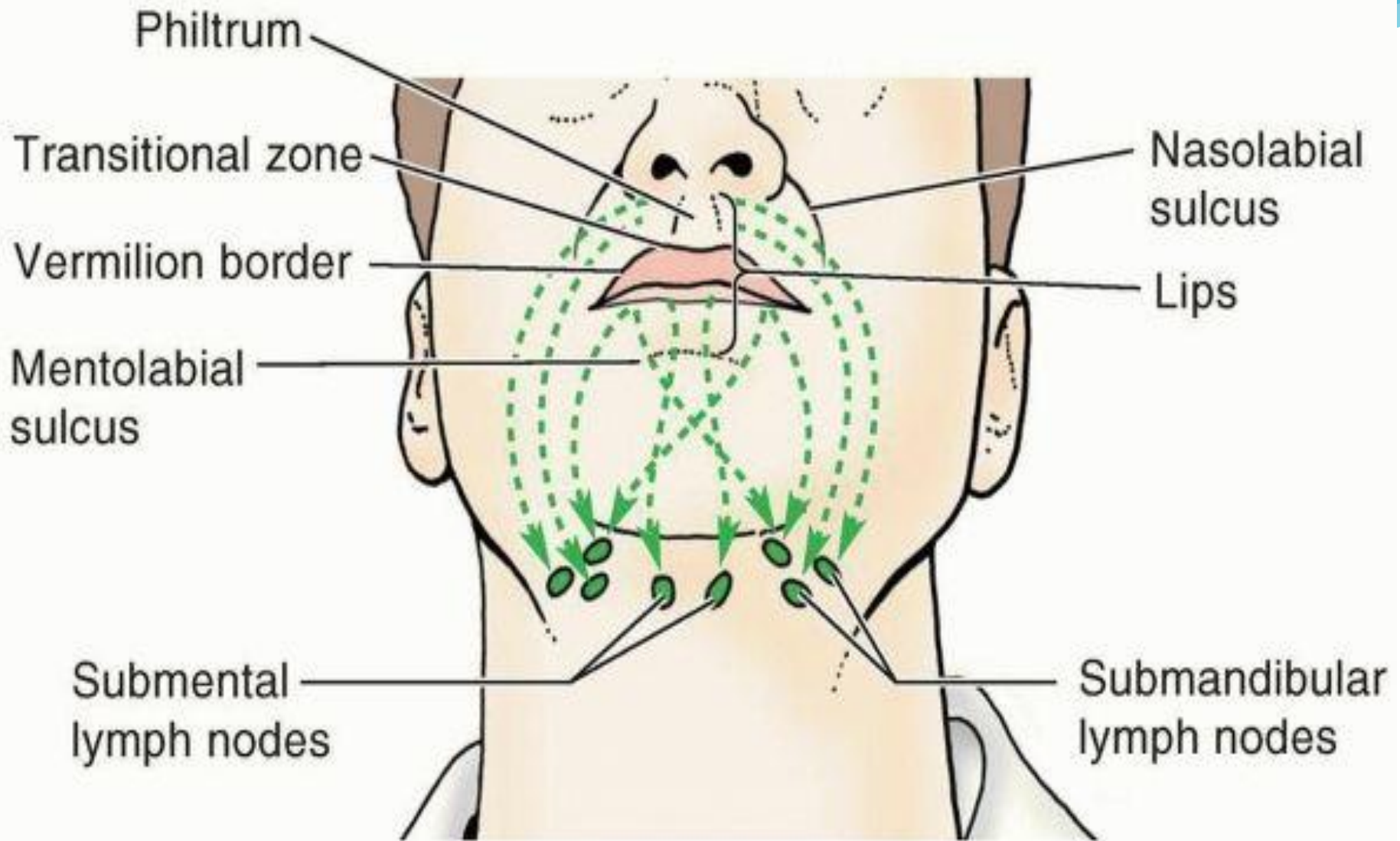
Alveolar mucosa
(unattached
gingiva)

(B)



- **Nerve supply:**
 - The **upper lip** is supplied by:
 - the **superior labial** branches of the **infra-orbital nerves** (**of CN V2**)
 - The **lower lip** is supplied by:
 - the inferior labial branches of the mental nerves (**of CN V3**)
- **Lymphatic drainage:**
 - lymph from the upper lip and lateral parts of the lower lip passes primarily to the **submandibular lymph nodes**
 - whereas lymph from the medial part of the lower lip passes initially to the **submental lymph nodes**.

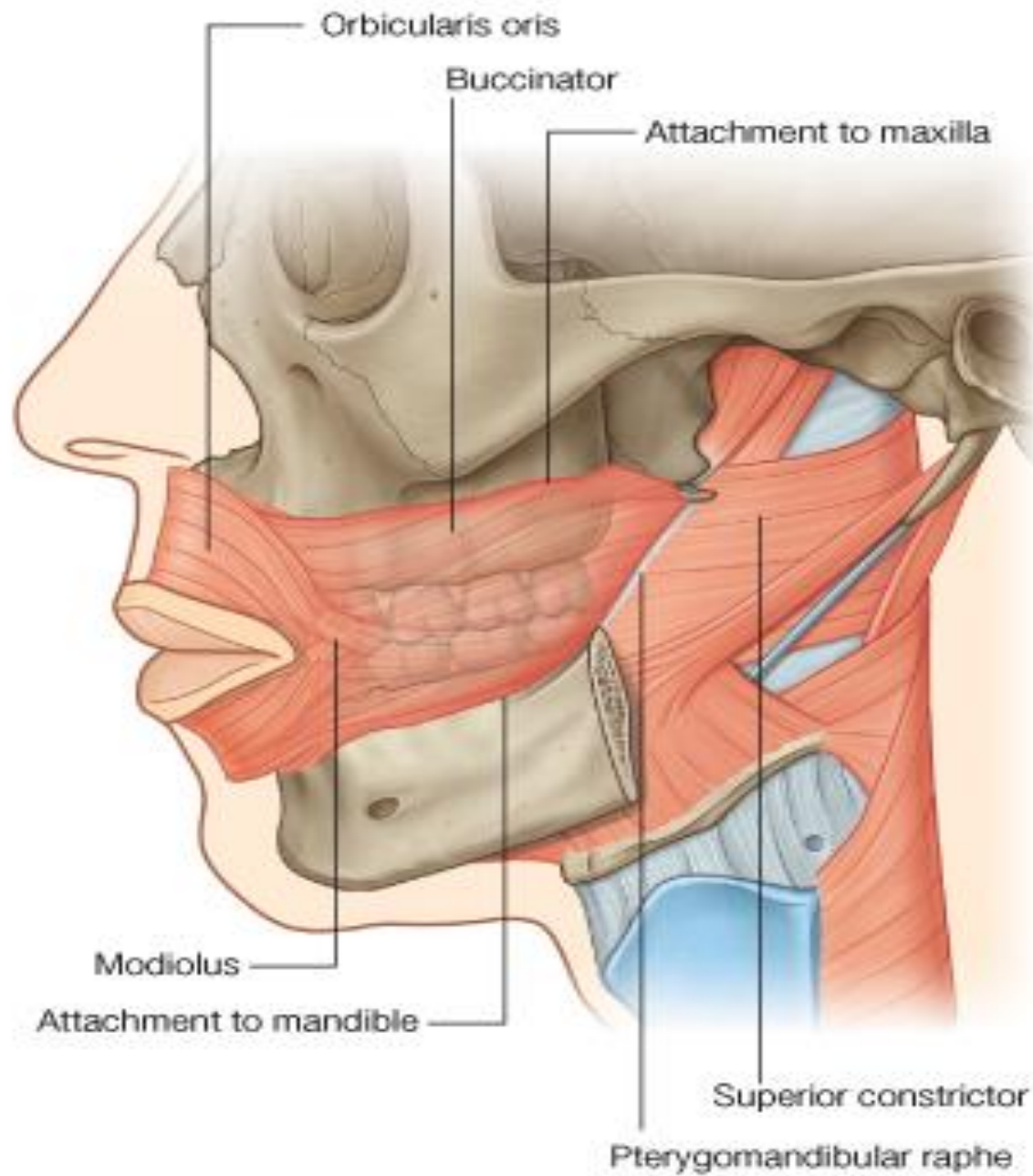




Anterior view

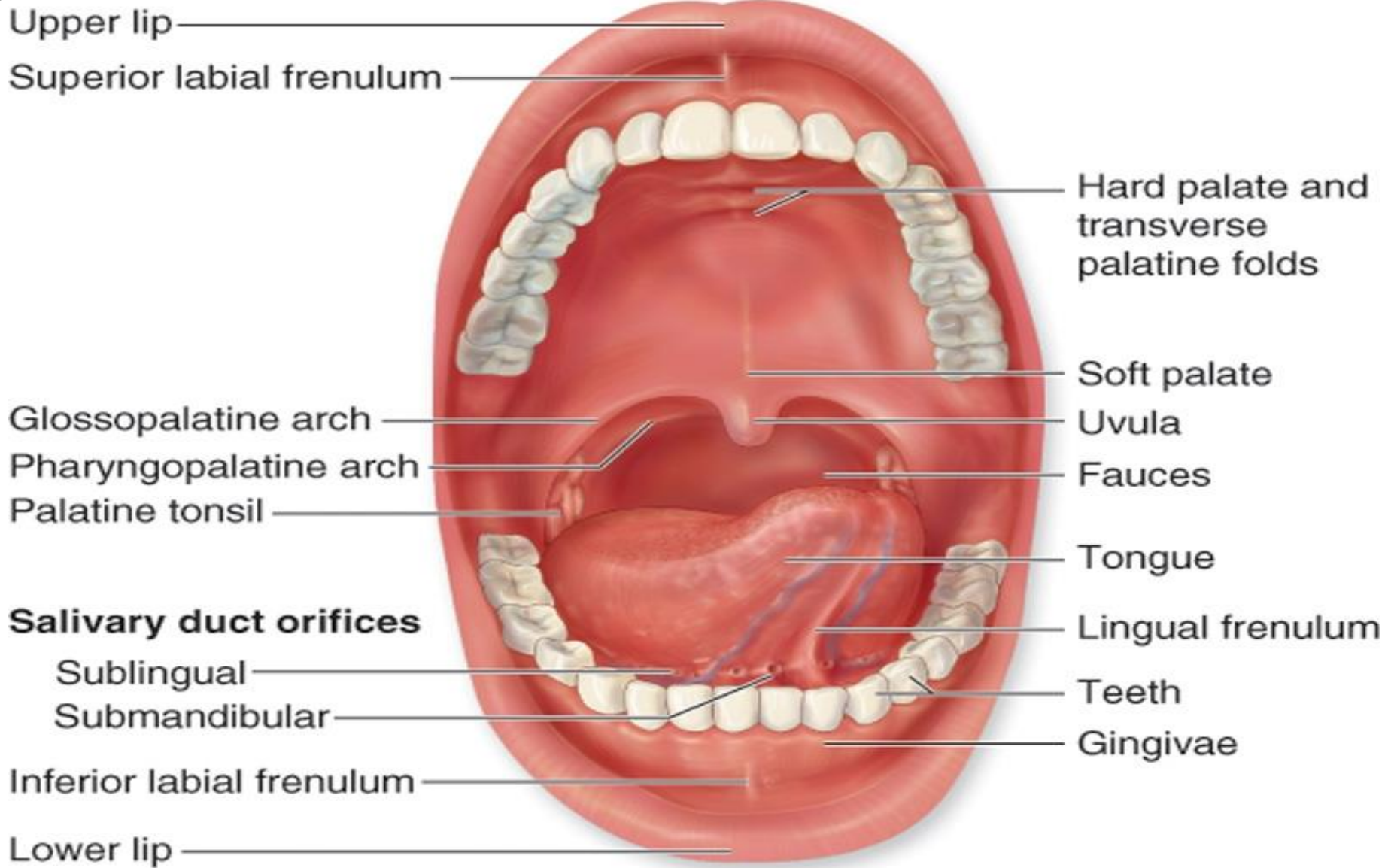
- The cheeks

- Anatomically, the external aspect of the cheeks constitutes the buccal region bounded:
 - anteriorly by the oral and mental regions (lips and chin)
 - superiorly by the zygomatic region
 - posteriorly by the parotid region
 - inferiorly by the inferior border of the mandible
- The prominence of the cheek occurs at the junction of the zygomatic and buccal regions



• The oral cavity proper

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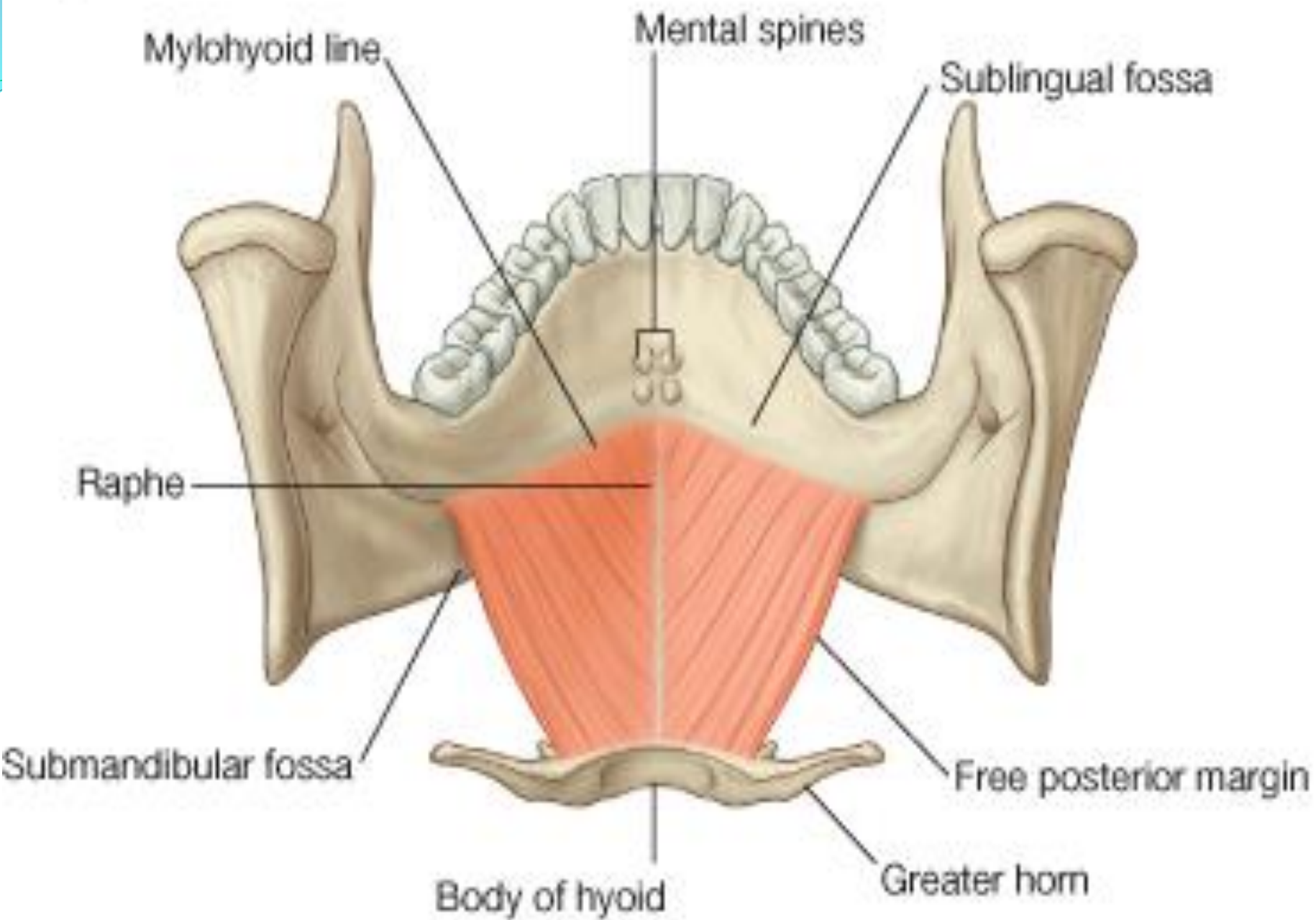
(a)

b. The oral cavity proper

- the space between the upper and the lower **dental arches** or arcades (maxillary and mandibular alveolar arches and the teeth they bear)
- It is limited laterally and anteriorly by the **dental arches**
- **The roof** of the oral cavity is formed by the **palate**
- **Posteriorly**, the oral cavity communicates with the **oropharynx** (oral part of the pharynx)
- When the mouth is closed and at rest, the oral cavity is fully occupied by the tongue

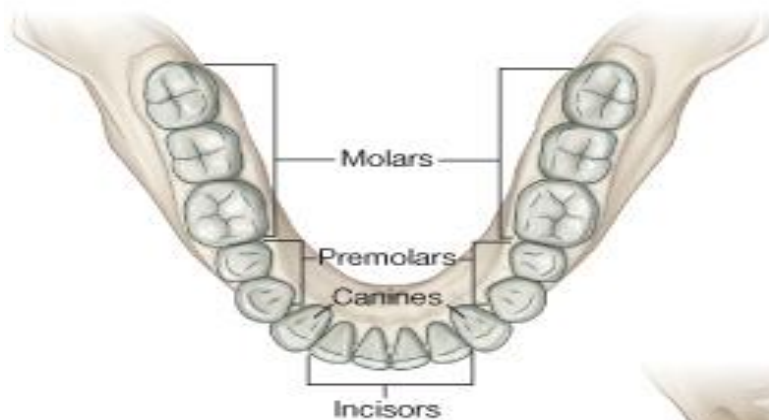
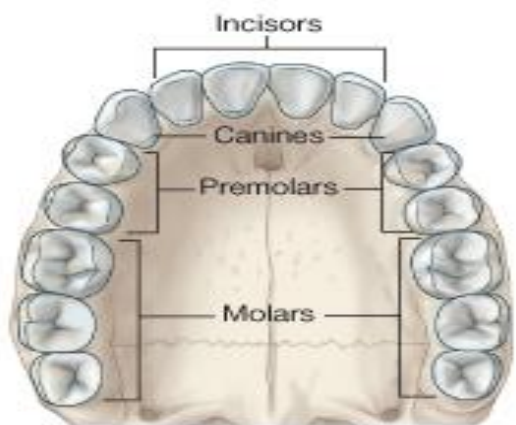
- The floor of the oral cavity proper:

- is formed mainly by three structures
 - a muscular diaphragm, which fills the U-shaped gap between the left and right sides of the body of the mandible and is composed of the paired mylohyoid muscles
 - two cord-like geniohyoid muscles above the diaphragm, which run from the mandible in front to the hyoid bone behind
 - the tongue, which is superior to the geniohyoid muscles.



Teeth:

- are attached to sockets (alveoli) in two elevated arches of bone on the mandible below and the maxillae above (alveolar arches)
- If the teeth are removed, the alveolar bone is resorbed and the arches disappear
- The **gingivae (gums)**:
 - are specialized regions of the oral mucosa that surround the teeth and cover adjacent regions of the alveolar bone.
- The different types of teeth are distinguished on the basis of morphology, position, and function

A**Upper**

Roots related to
maxillary sinus

Maxillary sinus

Lower

Roots related to
mandibular canal

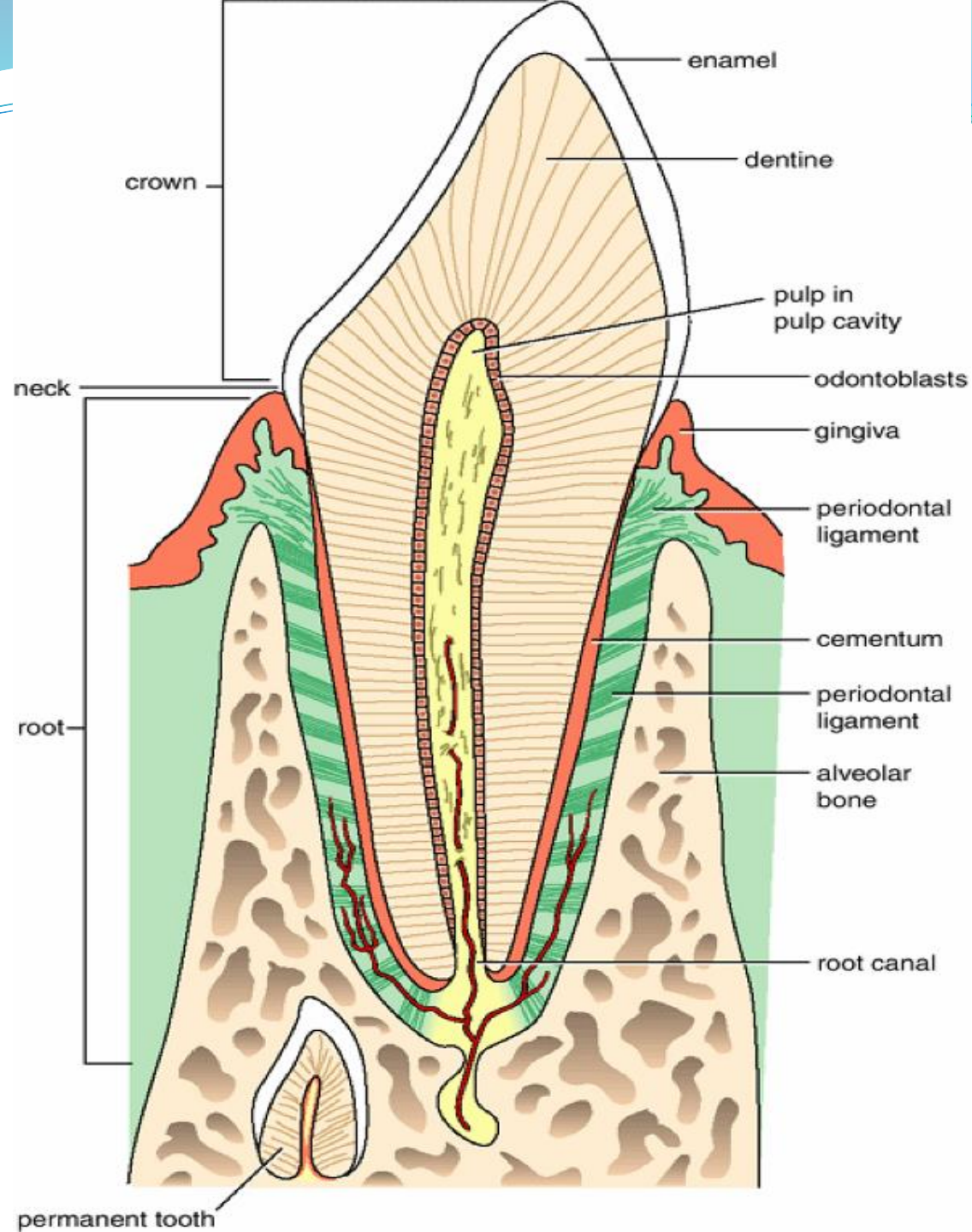
B**Upper**

Incisors

Lower

Canines

Molars





The chief functions of the teeth are to:

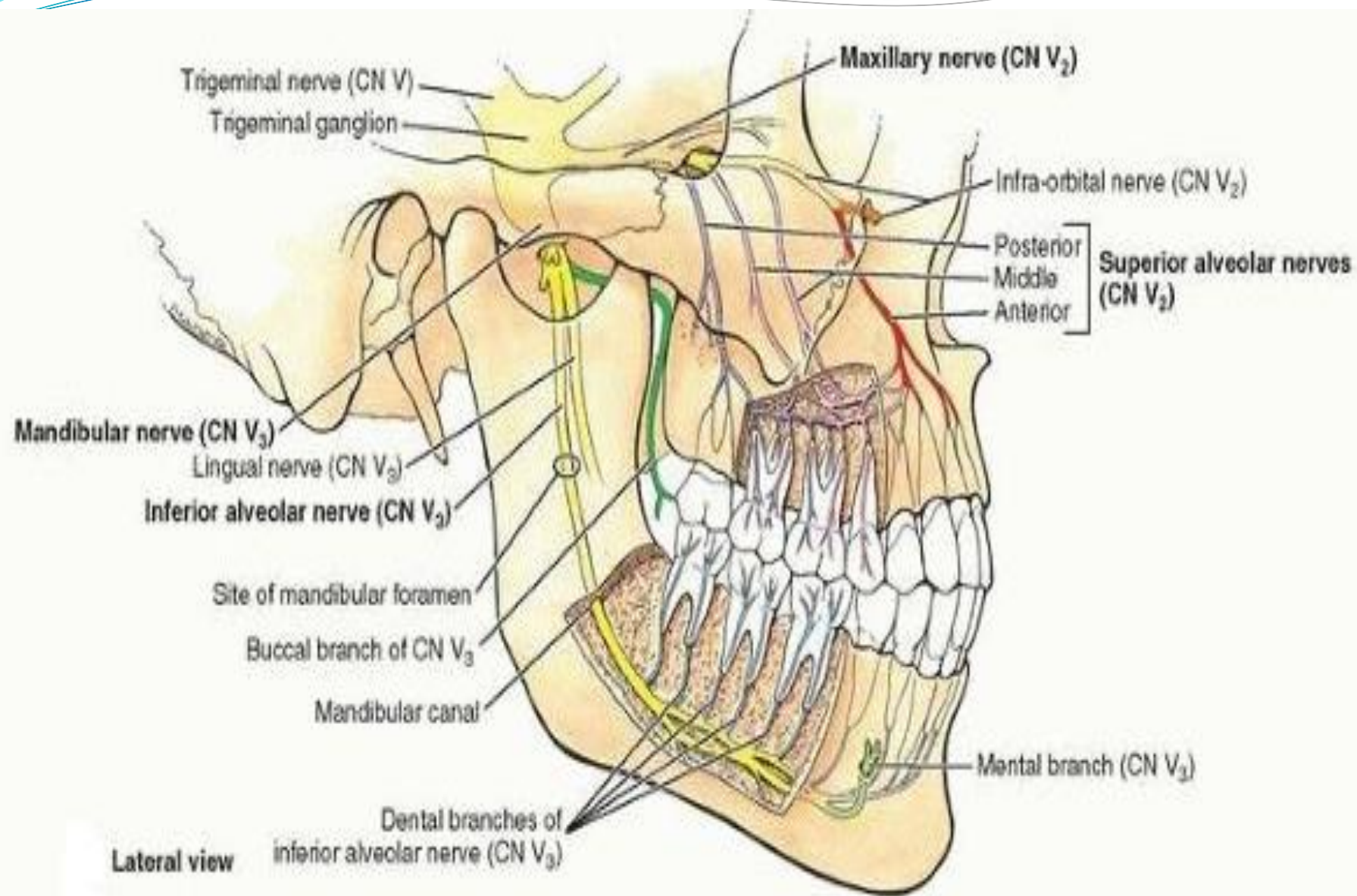
- **Incise, reduce, and mix food** material with saliva during mastication (chewing).
- Help sustain themselves in the tooth sockets by assisting the development and protection of the tissues that support them.
- Participate in articulation (distinct connected speech)

Deciduous Teeth	Central Incisor	Lateral Incisor	Canine	1st Molar	2nd Molar
Eruption (months) ^a	6–8	8–10	16–20	12–16	20–24
Shedding (years)	6–7	7–8	10–12	9–11	10–12

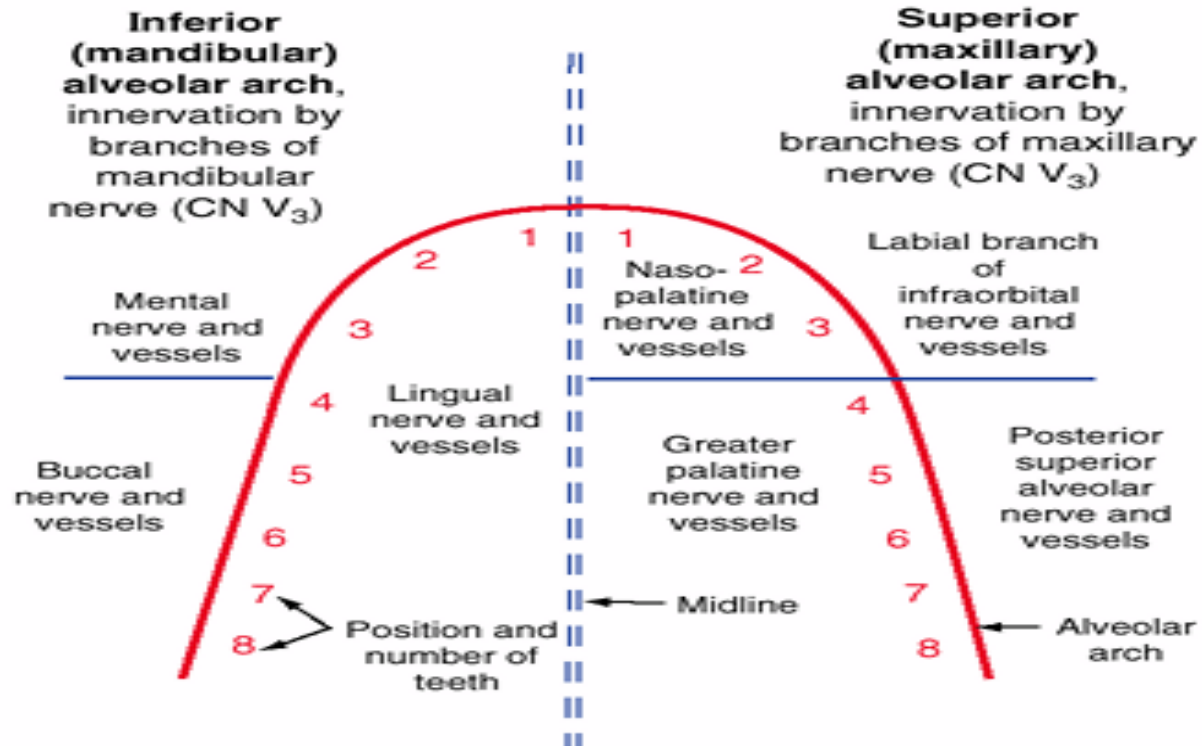
^aIn some normal infants, the first teeth (medial incisors) may not erupt until 12–13 months of age.

Permanent Teeth	Central Incisor	Lateral Incisor	Canine	1st Premolar	2nd Premolar	1st Molar	2nd Molar	3rd Molar
Eruption (years)	7–8	8–9	10–12	10–11	11–12	6–7	12	13–25

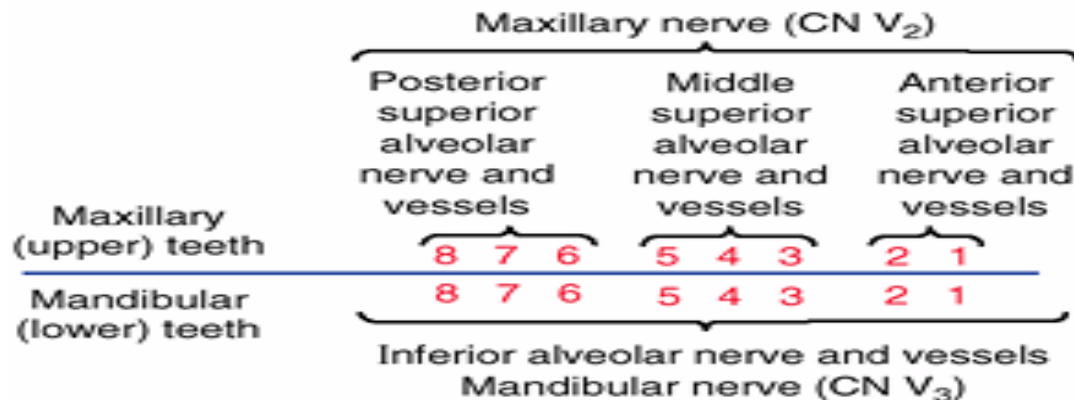
Innervations of teeth







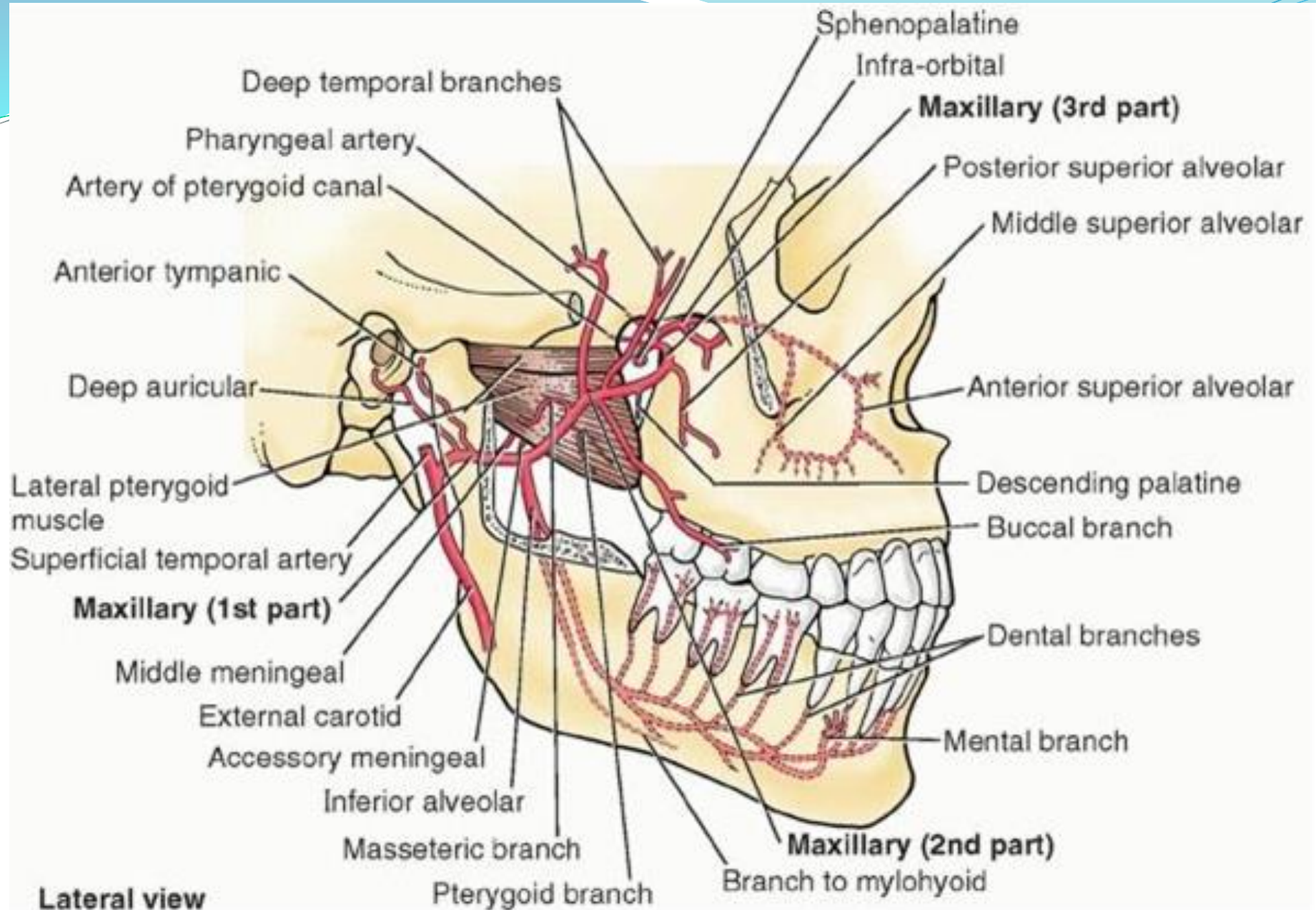
Nerves and vessels of gingiva, alveolar bone, and periodontium



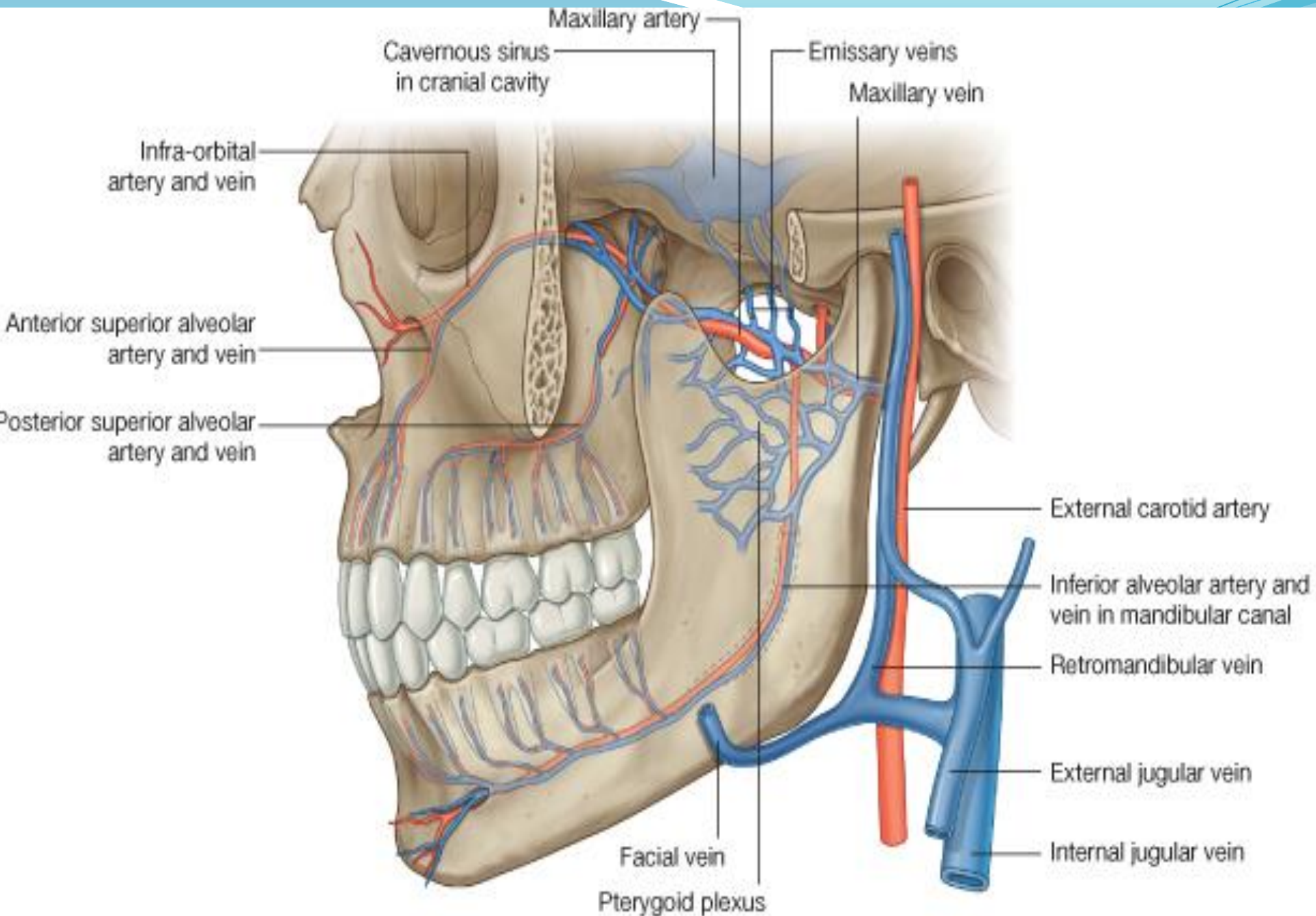
Nerves and vessels of teeth

VASCULATURE OF TEETH

- The **superior** and **inferior alveolar arteries**, branches of the maxillary artery, supply the maxillary and mandibular teeth
- Alveolar veins with the same names and distribution accompany the arteries.
- **Lymphatic vessels** from the teeth and gingivae pass mainly to the **submandibular lymph nodes**



Parts and branches of maxillary artery.



PALATE:

1. HARD PALATE

- is vaulted (concave)
- mostly filled by the tongue when it is at rest.
- The anterior two thirds of the palate has a bony skeleton formed by the **palatine processes** of the maxillae and **the horizontal plates** of the palatine bones
- The **incisive fossa** is a depression in the midline of the bony palate posterior to the central incisor teeth into which the incisive canals open.
- **The nasopalatine nerves pass** from the nose through a variable number of incisive canals and foramina that open into the incisive fossa



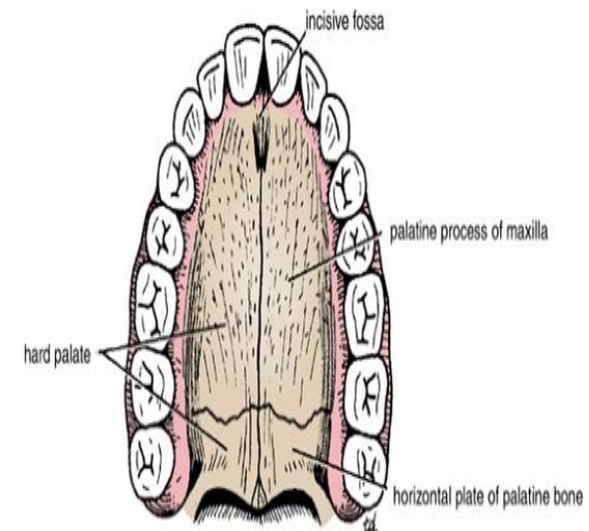
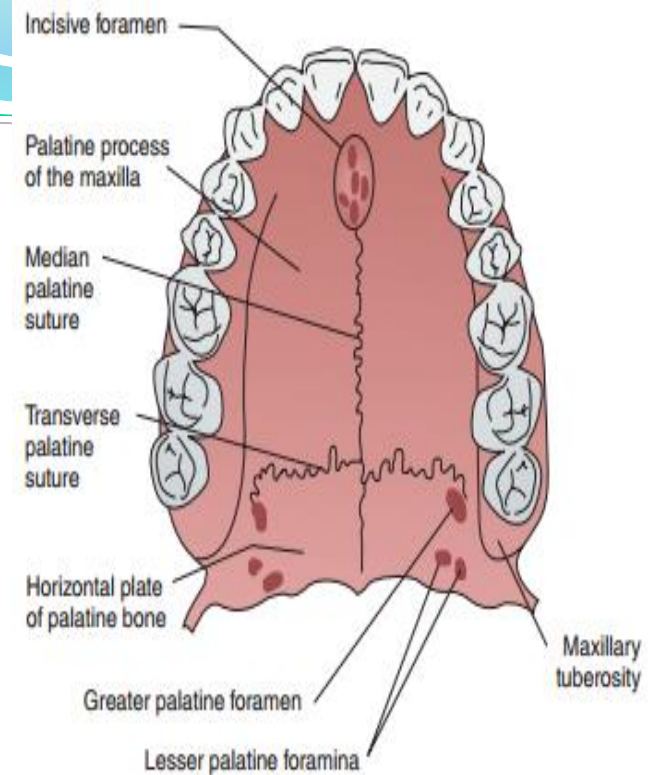
Incisive
papilla

Palatine
rugae

Palatine
raphe

(B)

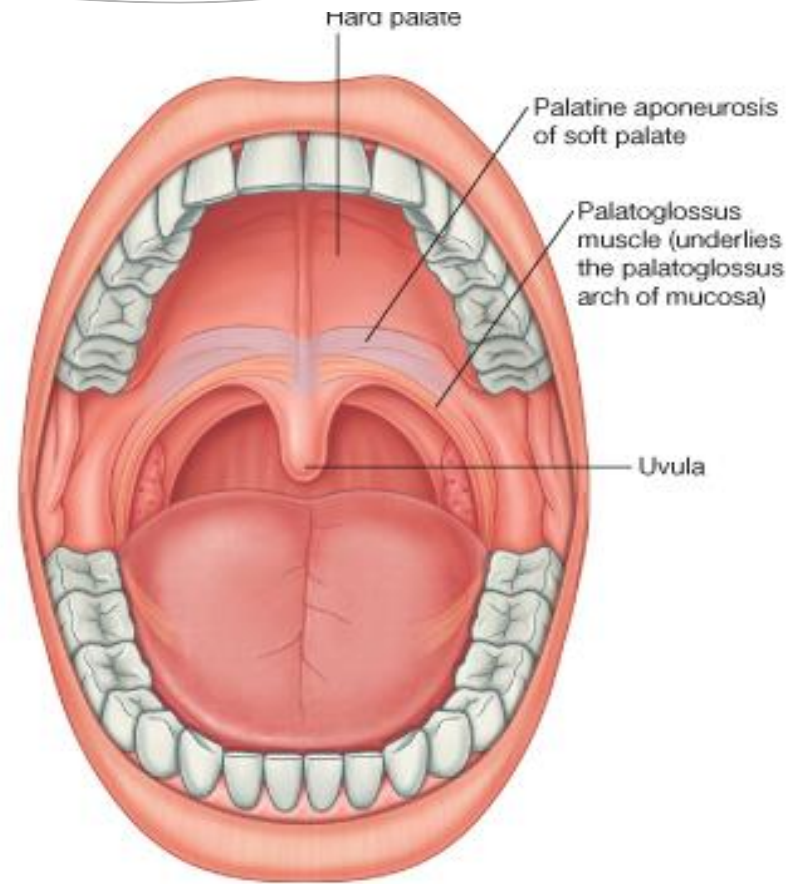
- Medial to the 3rd molar tooth, the greater palatine foramen pierces the lateral border of the bony palate
- The greater palatine vessels and nerve emerge from this foramen and run anteriorly on the palate.
- The lesser palatine foramina posterior to the greater palatine foramen pierce the pyramidal process of the palatine bone.
- These foramina transmit the lesser palatine nerves and vessels to the soft palate and adjacent structures



Soft Palate

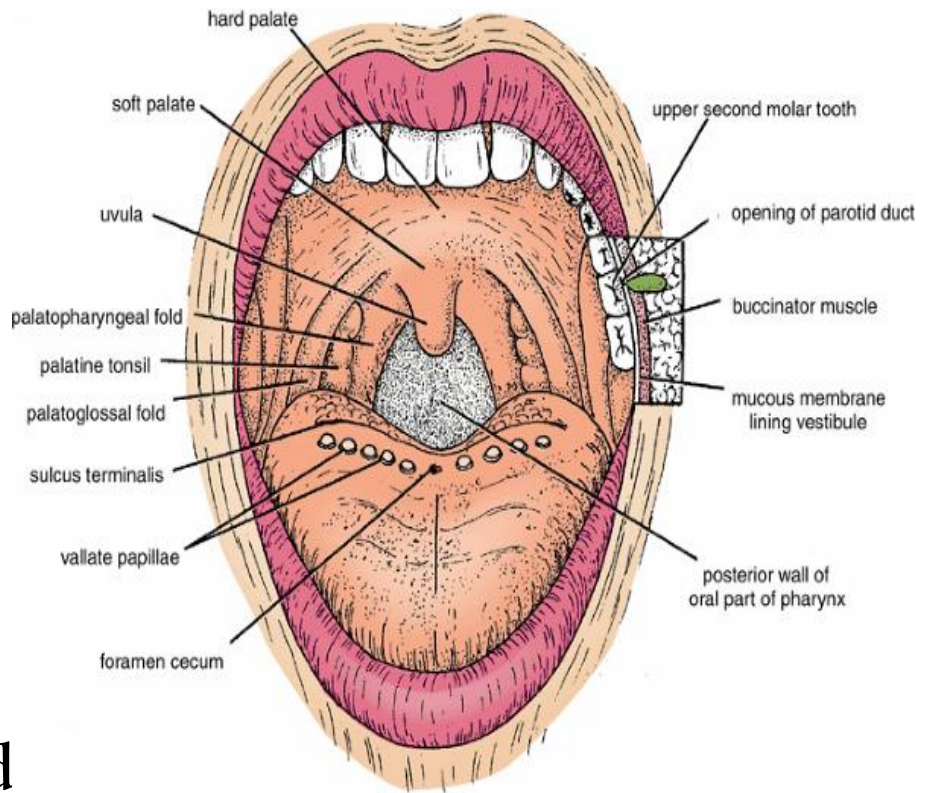
- is the movable **posterior third** of the palate
- is suspended from the posterior border of the hard palate
- strengthened by the **palatine aponeurosis**, which attaches to the posterior edge of the hard palate.
- The **aponeurosis**
 - **is thick anteriorly** and **thin posteriorly**, where it blends with a **posterior muscular part**.
 - formed by the expanded tendon of the **tensor veli palatini muscle**
- **Posteroinferiorly:**
 - the soft palate has a curved free margin from which hangs a conical process, **the uvula**

- Laterally, the soft palate is continuous with the wall of the pharynx and is joined to the tongue and pharynx by the palatoglossal and palatopharyngeal arches, respectively
- A few taste buds are located in the epithelium covering the oral surface of the soft palate



• The fauces (L. the throat):

- is the space between the cavity of the mouth and the pharynx
- is bounded :
 - superiorly by the soft palate
 - inferiorly by the root of the tongue
 - laterally by the pillar of the fauces, the palatoglossal and palatopharyngeal arches



The isthmus of the fauces is the **short constricted space** that establishes the connection between the **oral cavity proper** and **the oropharynx**

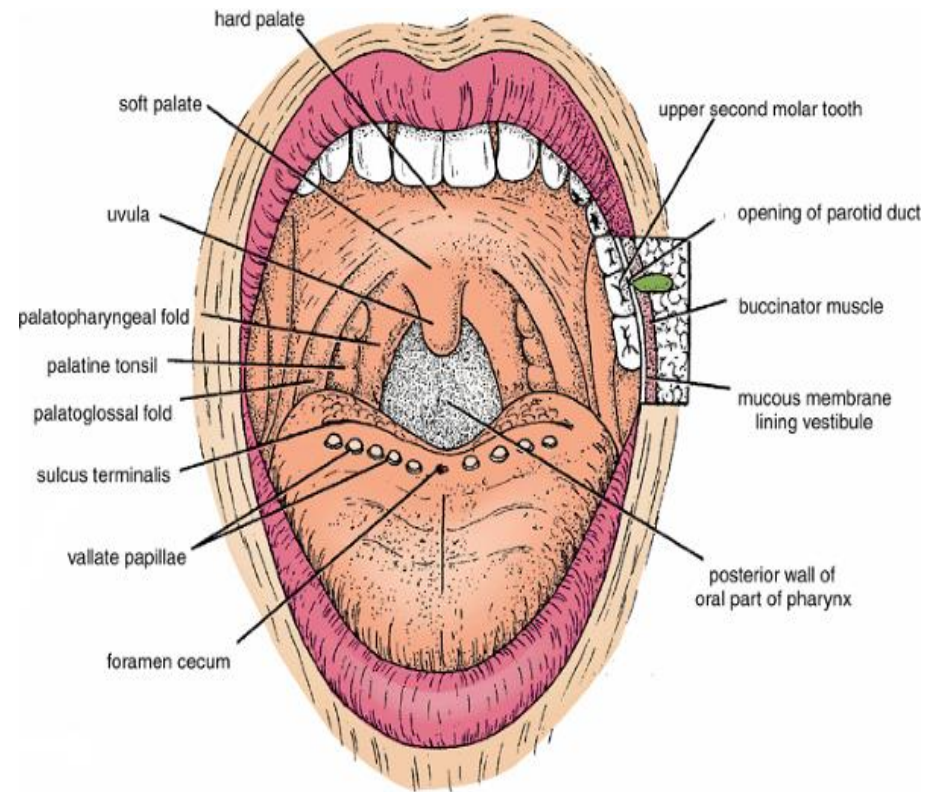
The isthmus is bounded:

anteriorly by the

palatoglossal folds

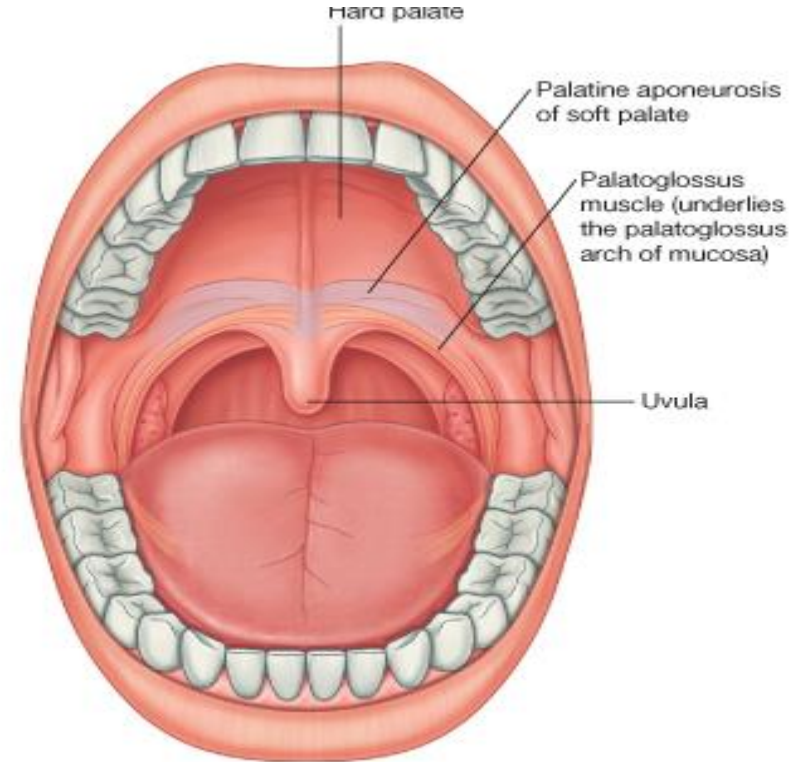
and posteriorly by the

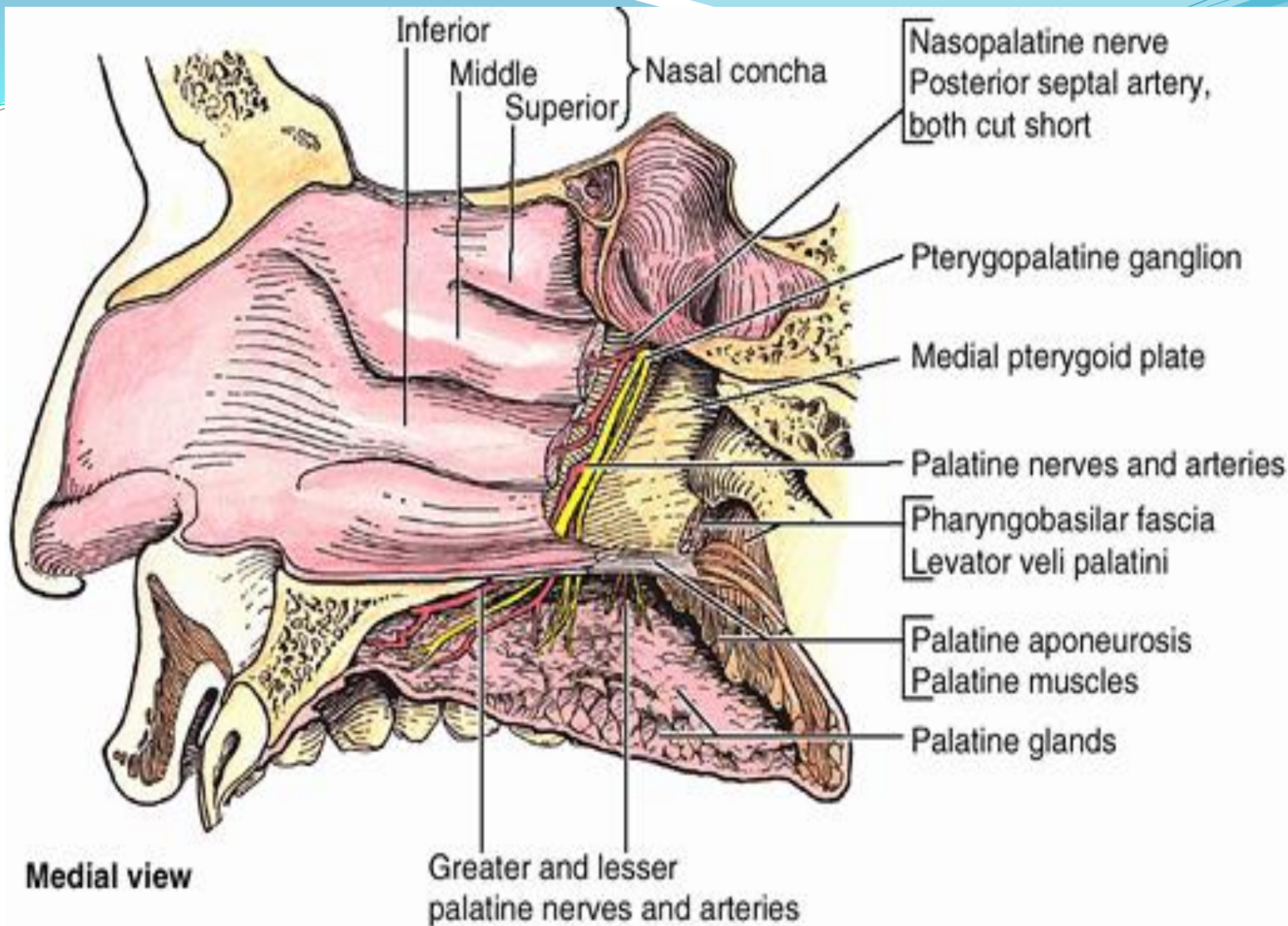
palatopharyngeal folds

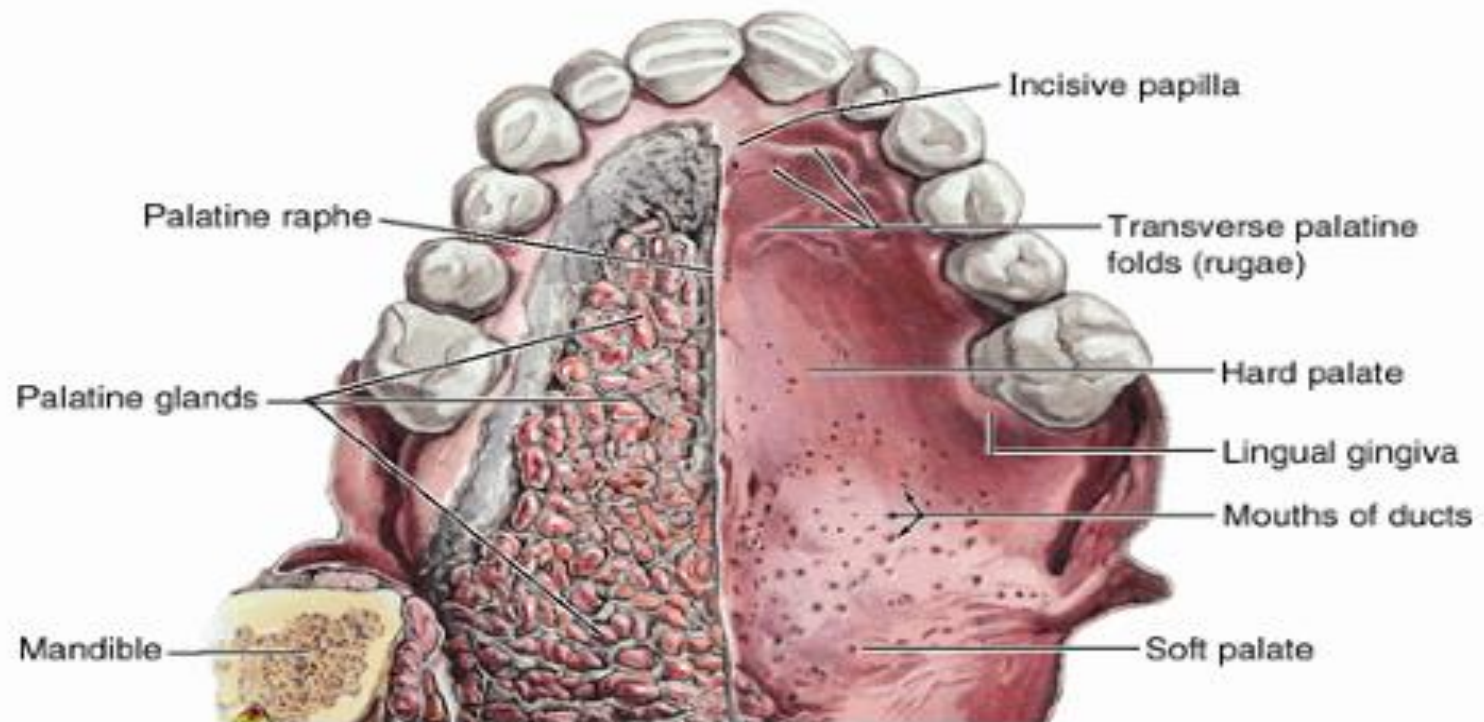


The palatine tonsils:

- are masses of lymphoid tissue, one on each side of the oropharynx
- Each tonsil is in a tonsillar sinus (fossa), bounded by the palatoglossal and palatopharyngeal arches and the tongue



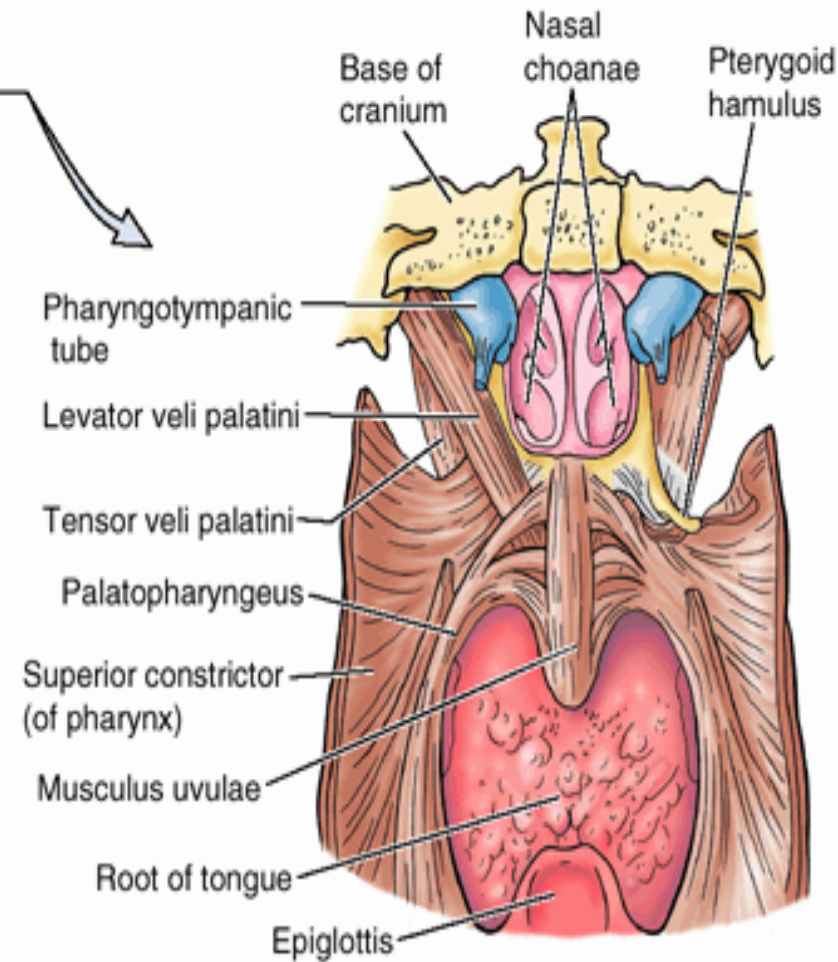
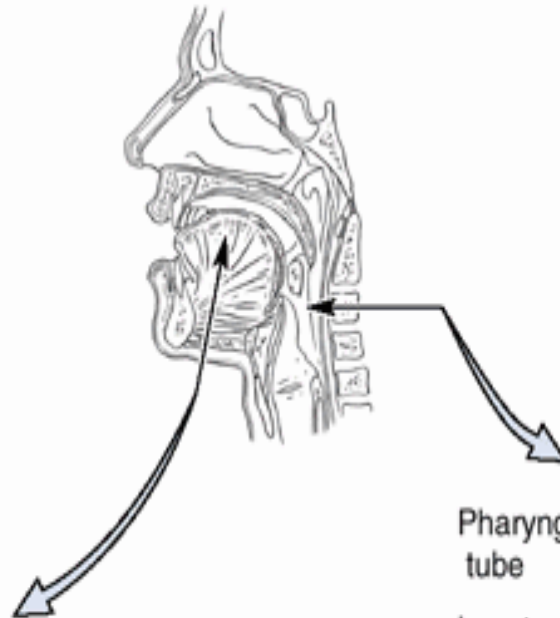
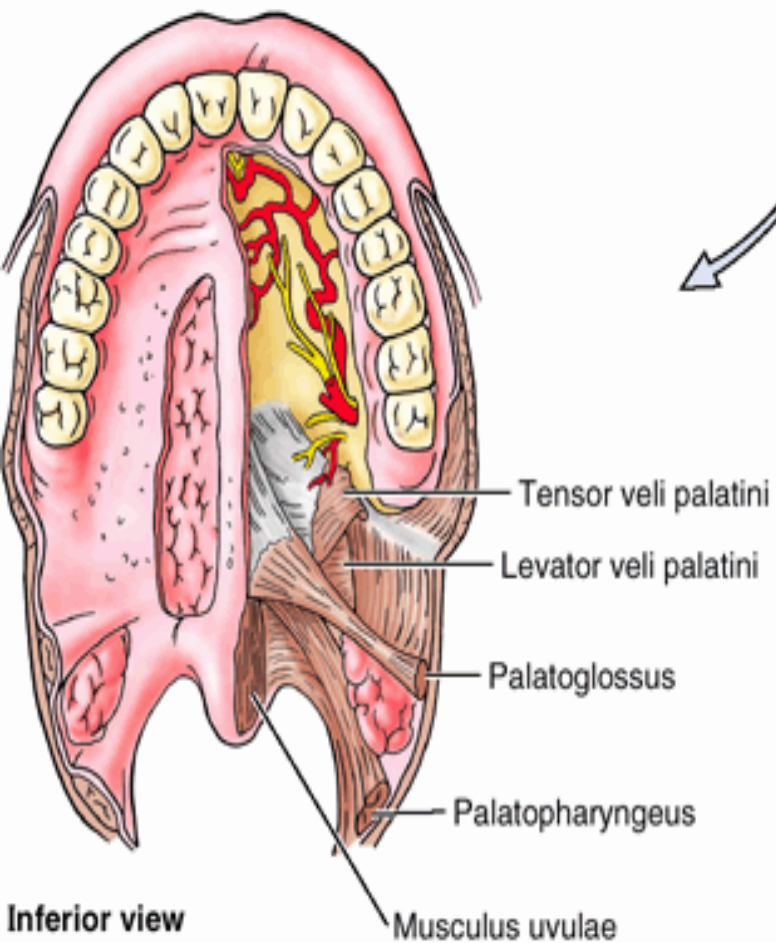




Muscles of the Soft Palate

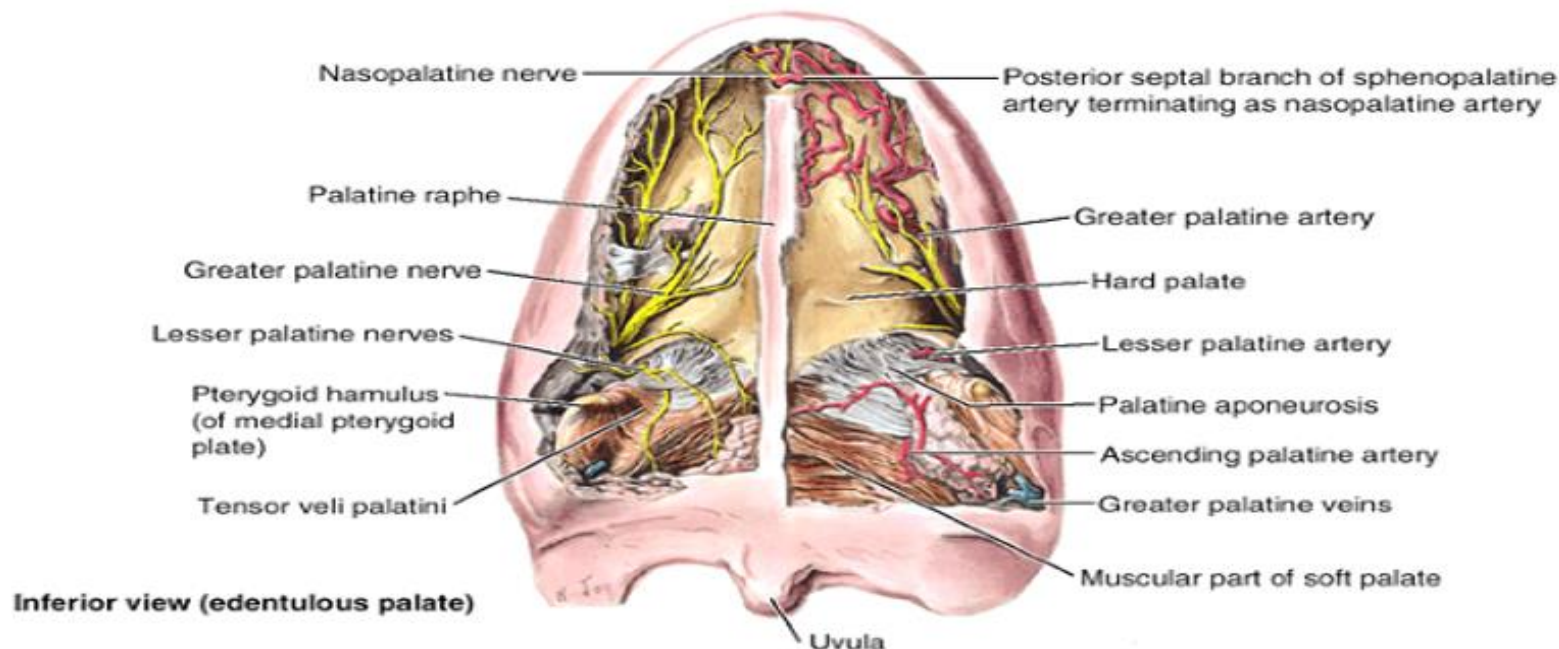
- The **five muscles** of the soft palate arise from the base of the cranium and descend to the palate
- The soft palate may be elevated so that it is in contact with the posterior wall of the pharynx
 - This closes the pharyngeal isthmus, requiring that one breathes through the mouth.
- The soft palate may also be drawn inferiorly so that it is in contact with the posterior part of the tongue.
 - This closes the isthmus of the fauces, so that expired air passes through the nose (even when the mouth is open) and prevents substances in the oral cavity from passing to the pharynx

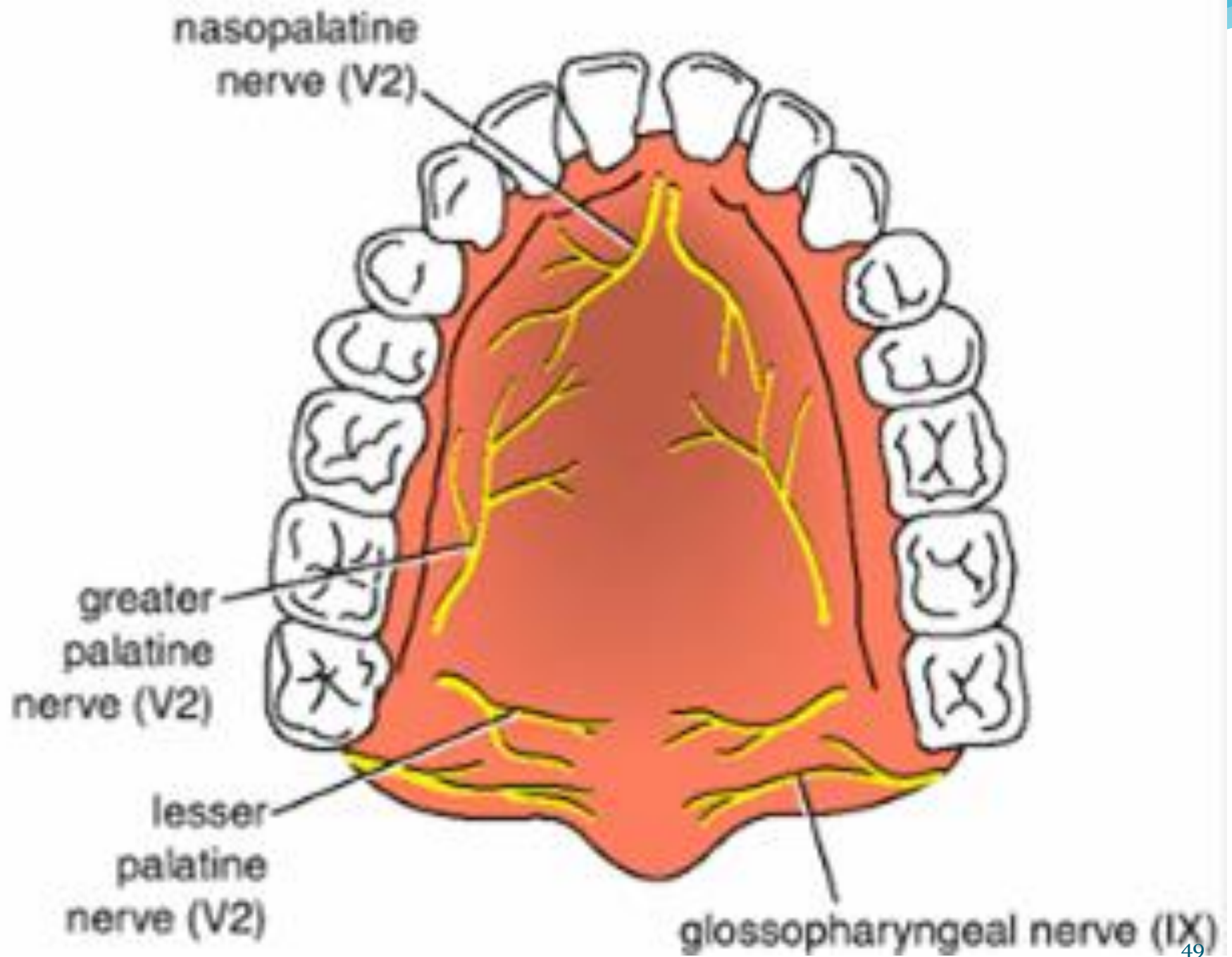
Muscle	Superior Attachment	Inferior Attachment	Innervation	Main Action
Tensor veli palatini	Scaphoid fossa of medial pterygoid plate, spine of sphenoid bone, and cartilage of pharyngotympanic tube	Palatine aponeurosis	Medial pterygoid nerve (a branch of mandibular nerve, CN V ₃) via otic ganglion	Tenses soft palate and opens mouth of pharyngotympanic tube during swallowing and yawning
Levator veli palatini	Cartilage of pharyngotympanic tube and petrous part of temporal bone		Pharyngeal branch of vagus nerve (CN X) via pharyngeal plexus	Elevates soft palate during swallowing and yawning
Palatoglossus	Palatine aponeurosis	Side of tongue		Elevates posterior part of tongue and draws soft palate onto tongue
Palatopharyngeus	Hard palate and palatine aponeurosis	Lateral wall of pharynx		Tenses soft palate and pulls walls of pharynx superiorly, anteriorly, and medially during swallowing
Musculus uvulae	Posterior nasal spine and palatine aponeurosis	Mucosa of uvula		Shortens uvula and pulls it superiorly



Nerve supply of the palate

- The **greater and lesser palatine nerves** from the maxillary division of the trigeminal nerve enter the palate through the **greater and lesser palatine foramina**
- The **nasopalatine nerve**, also a branch of the maxillary nerve, enters the front of the hard palate through the **incisive foramen**.
- The **glossopharyngeal nerve** also supplies the soft palate

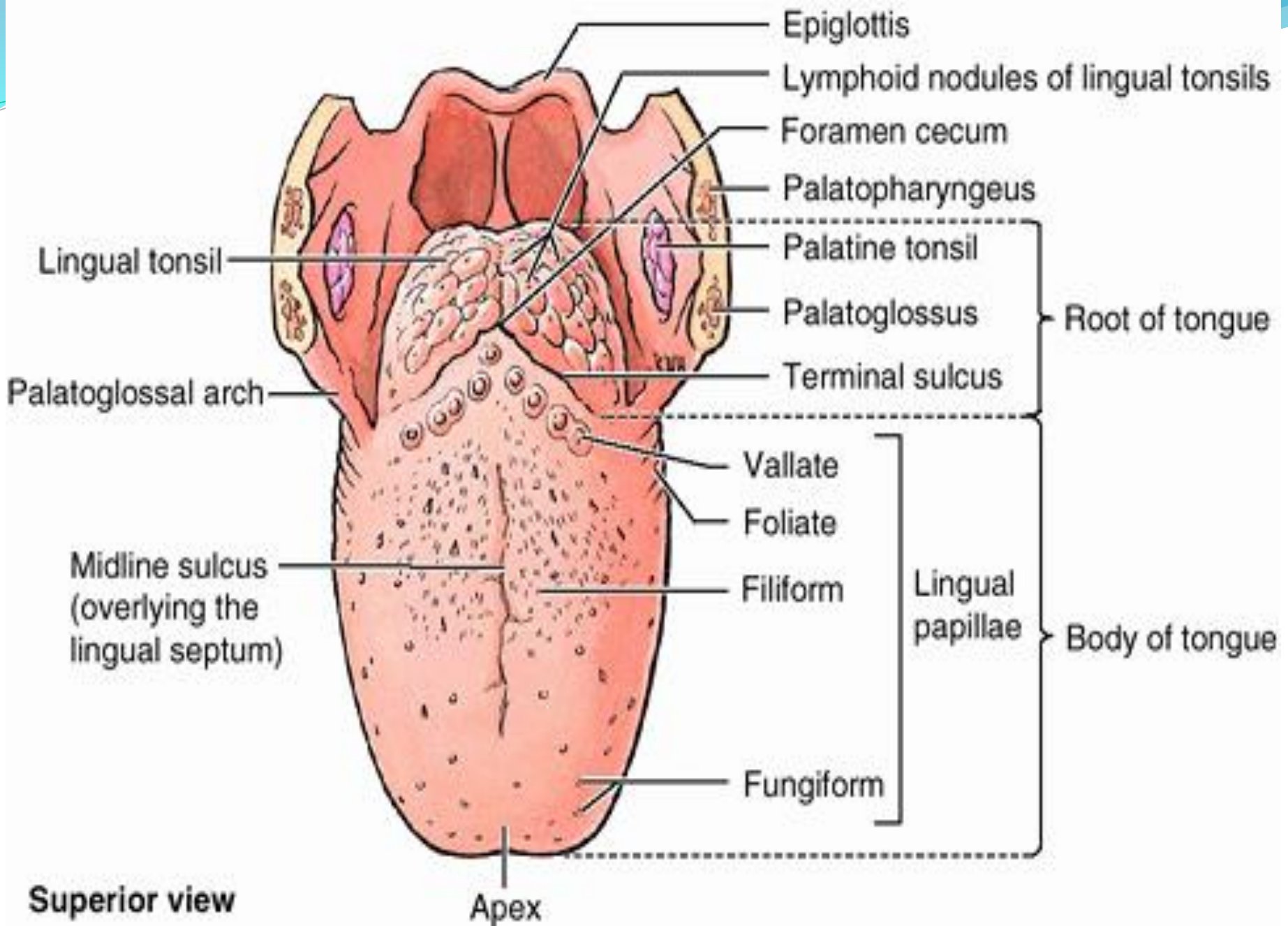


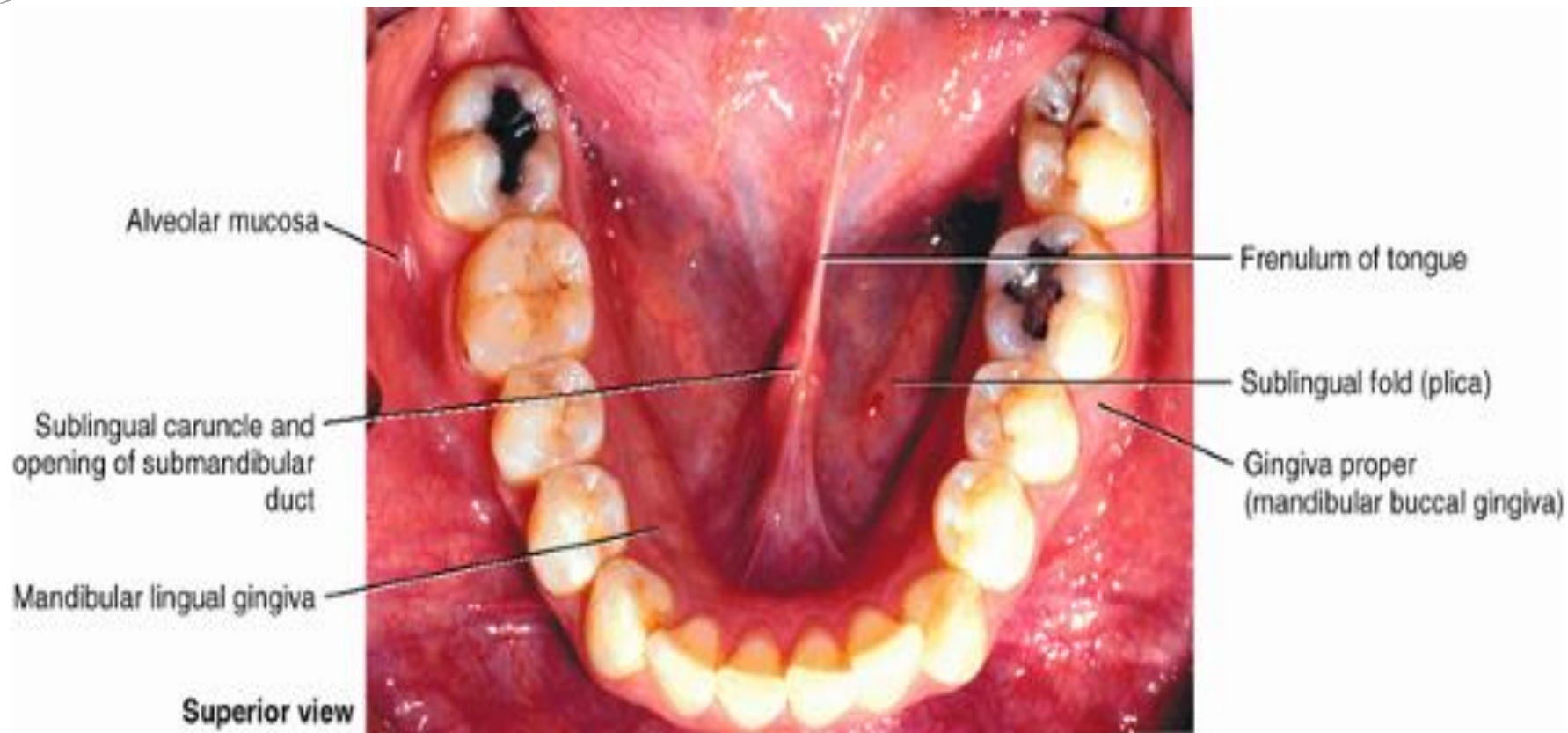


THE TONGUE

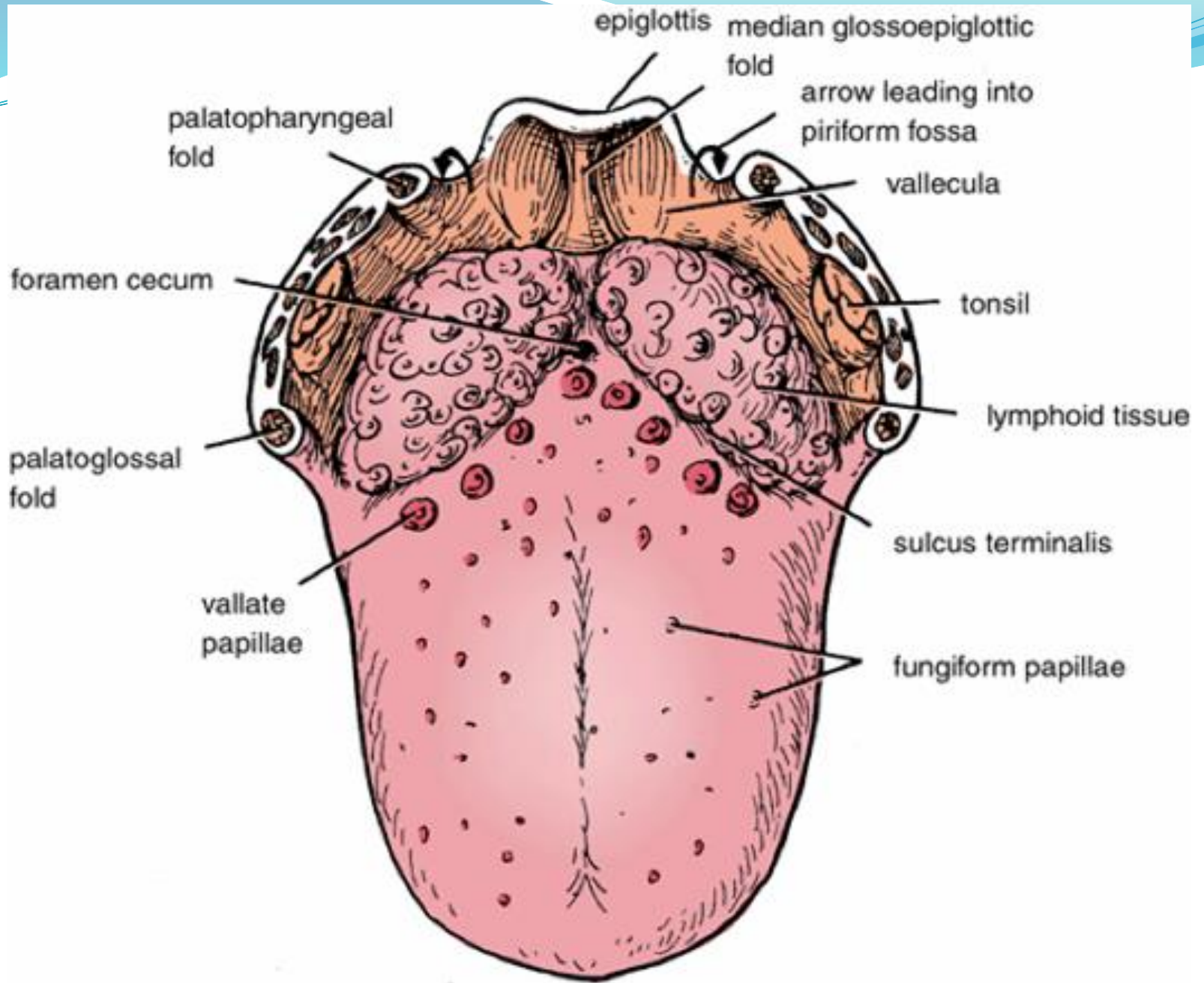
Parts and Surfaces of the Tongue

- The tongue has a root, a body, an apex, a curved dorsum, and an inferior surface
 - **The root of the tongue:**
 - is the part of the tongue that rests on the floor of the mouth
 - It is usually defined as the posterior third of the tongue.
 - **The body of the tongue:**
 - is the anterior two thirds of the tongue.
 - **The apex (tip) of the tongue:**
 - is the anterior end of the body, which rests against the incisor teeth.
 - **The body and apex** of the tongue are extremely mobile.





- A shallow midline groove of the tongue divides the tongue into right and left halves.
- The groove also indicates the site of fusion of the embryonic distal tongue buds
- The mucous membrane of the posterior part of the tongue is **thick and freely movable**
- It has no lingual papillae, but the underlying lymphoid nodules give this part of the tongue an irregular, **cobblestone appearance.**
- The lymphoid nodules are known collectively as **the lingual tonsil.**

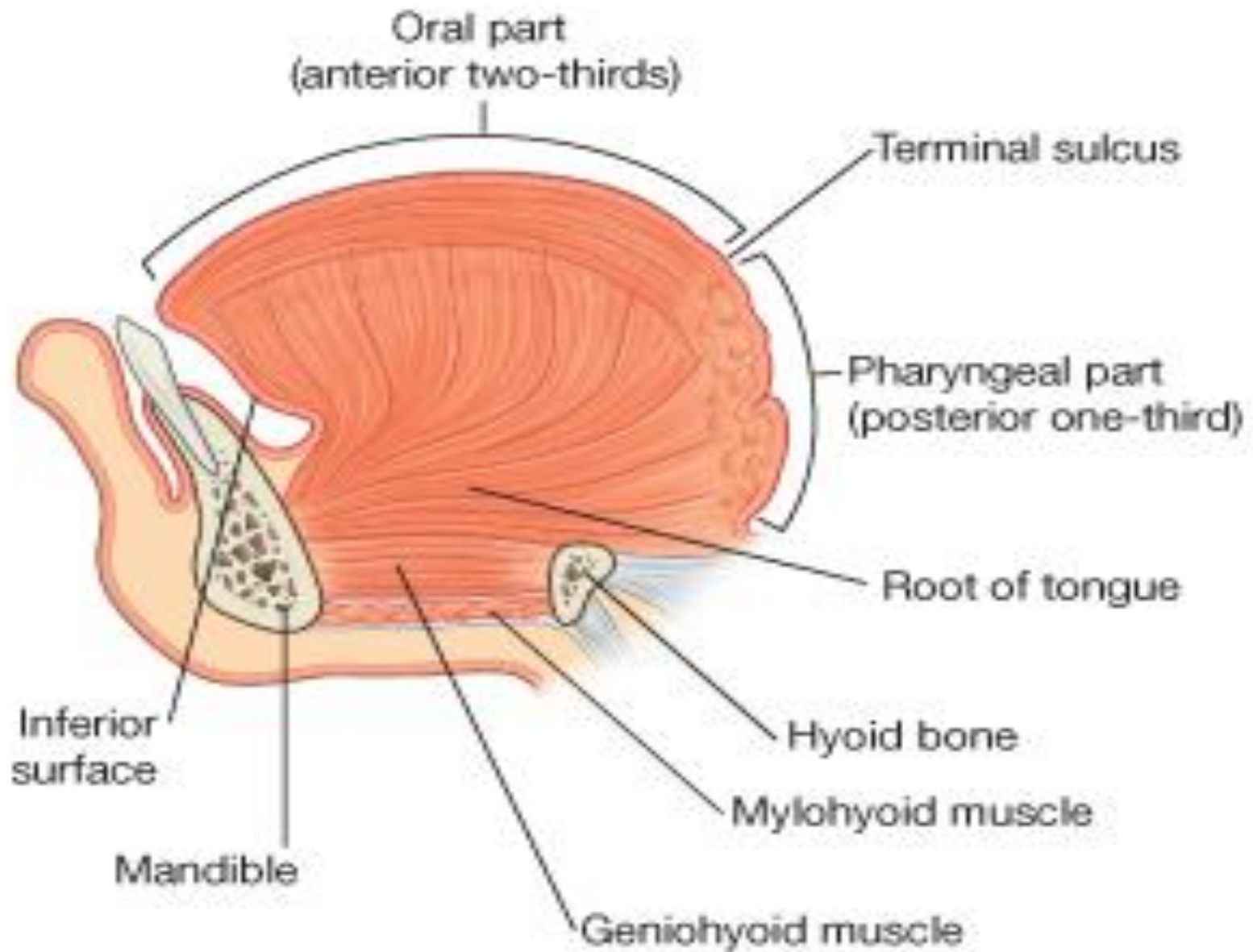


- The pharyngeal part of the tongue constitutes the anterior wall of the **oropharynx** and can be inspected only with a mirror or downward pressure on the tongue with a tongue depressor
- The inferior surface of the tongue:
 - is covered with a thin, transparent mucous membrane through which one can see the underlying veins.
 - This surface is connected to the floor of the mouth by a midline fold called the **frenulum of the tongue**
 - The frenulum allows the anterior part of the tongue to move freely.
 - **On each side of the frenulum**, a deep lingual vein is visible through the thin mucous membrane
 - **A sublingual caruncle (papilla)** is present on each side of the base of the **lingual frenulum** that includes the opening of the **submandibular duct** from the **submandibular** salivary gland.

Muscles of the Tongue

- The four intrinsic and four extrinsic muscles in each half of the tongue are separated by a median fibrous lingual septum, which merges posteriorly with the lingual aponeurosis
- **Extrinsic Muscles of the Tongue**
 - Are genioglossus, hyoglossus, styloglossus, and palatoglossus
 - originate outside the tongue and attach to it.
 - They mainly move the tongue but they can alter its shape as well

Muscle	Shape and Position	Proximal Attachment	Distal Attachment	Main Action(s)
Extrinsic muscles of the tongue				
Genioglossus	Fan-shaped muscle; constitutes bulk of tongue	Via a short tendon from superior part of mental spine of mandible	Entire dorsum of tongue; inferior most and posterior most fibers attach to body of hyoid bone	Bilateral activity depresses tongue, especially central part, creating a longitudinal furrow; posterior part pulls tongue anteriorly for protrusion; ^a most anterior part retracts apex of protruded tongue; unilateral contraction deviates (â€œwagsâ€œ) tongue to contralateral side
Hyoglossus	Thin, quadrilateral muscle	Body and greater horn of hyoid bone	Inferior aspects of lateral part of tongue	Depresses tongue, especially pulling its sides inferiorly; helps shorten (retrude) tongue
Styloglossus	Small, short triangular muscle	Anterior border of distal styloid process; stylohyoid ligament	Sides of tongue posteriorly, interdigitating with hyoglossus	Retrudes tongue and curls (elevates) its sides, working with genioglossus to form a central trough during swallowing



Palatoglossus	Narrow crescent-shaped palatine muscle; forms posterior column of isthmus of fauces	Palatine aponeurosis of soft palate	Enters posterolateral tongue transversely, blending with intrinsic transverse muscles	Capable of elevating posterior tongue or depressing soft palate; most commonly acts to constrict isthmus of fauces
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Intrinsic Muscles of the Tongue

- The superior and inferior longitudinal
- transverse
- and vertical muscles are confined to the tongue.
- They have their attachments entirely within the tongue and are not attached to bone.
- **The superior and inferior longitudinal muscles:**
 - act together to make the tongue **short and thick** and to **retract the protruded tongue**.
- **The transverse and vertical muscles:**
 - act simultaneously to make the tongue long and narrow, which may push the tongue against the incisor teeth or protrude the tongue from the open mouth (especially when acting with the posterior inferior part of the genioglossus).

Intrinsic muscles of tongue

Superior longitudinal	Thin layer deep to mucous membrane of dorsum	Submucous fibrous layer and median fibrous septum	Margins of tongue and mucous membrane	Curls tongue longitudinally upward, elevating apex and sides of tongue; shortens (retrudes) tongue
Inferior longitudinal	Narrow band close to inferior surface	Root of tongue and body of hyoid bone	Apex of tongue	Curls tongue longitudinally downward, depressing apex; shortens (retrudes) tongue
Transverse	Deep to superior longitudinal muscle	Median fibrous septum	Fibrous tissue at lateral lingual margins	Narrows and elongates (protrudes) tongue ^a
Vertical	Fibers intersect transverse muscle	Submucous fibrous layer of dorsum of tongue	Inferior surface of borders of tongue	Flattens and broadens tongue ^a

**Extrinsic
muscles**

Palatoglossus

Styloglossus

Hyoglossus

Genioglossus

Superior longitudinal

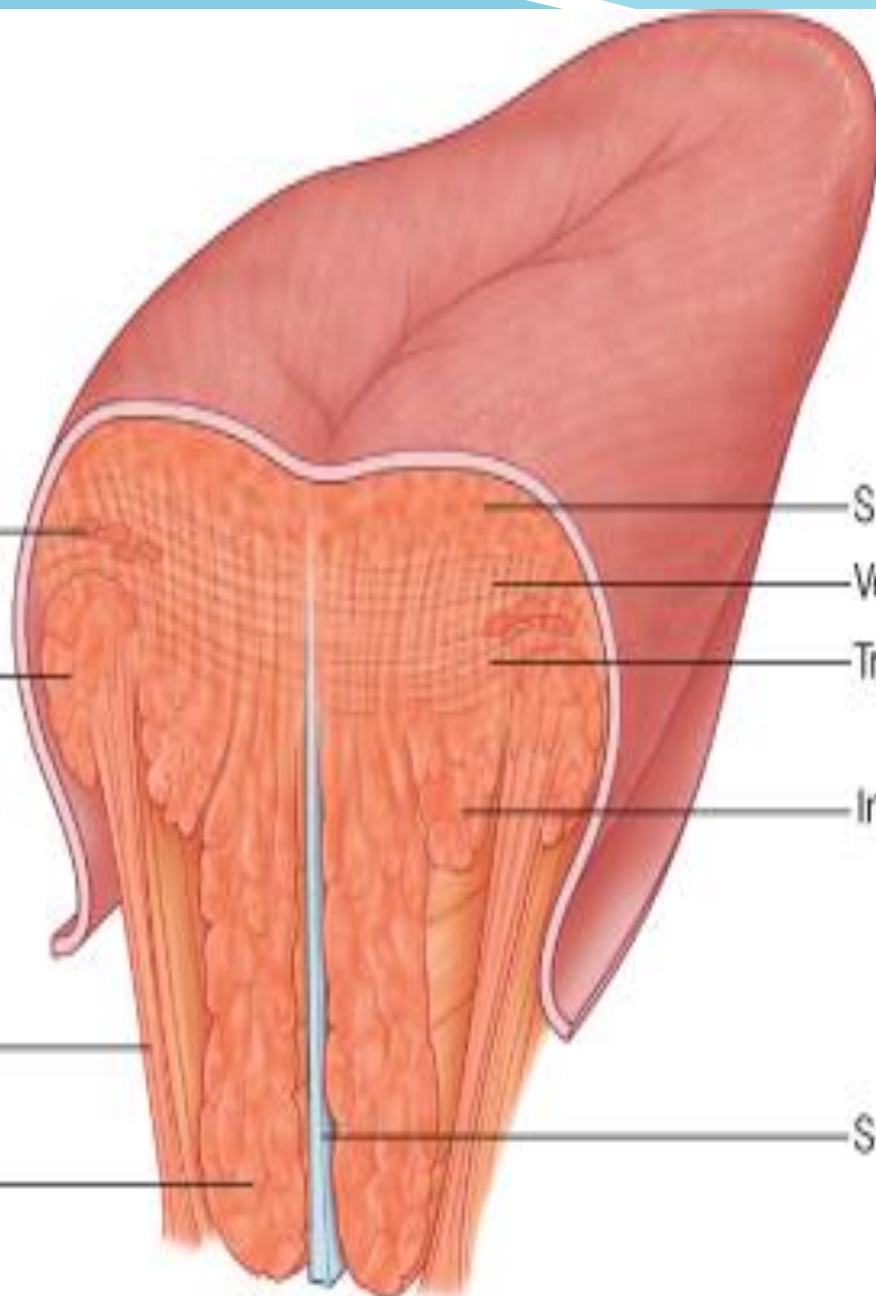
Vertical

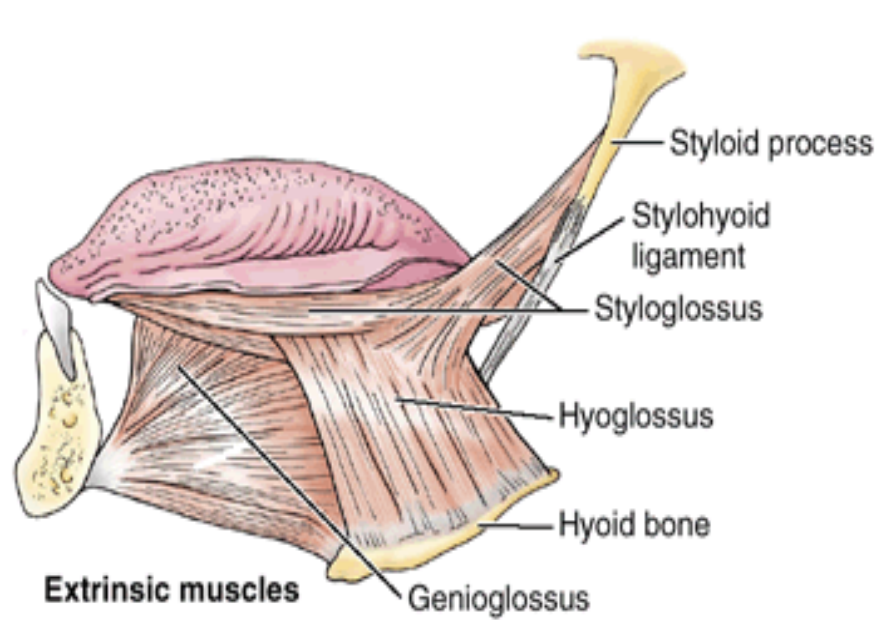
Transverse

Inferior longitudinal

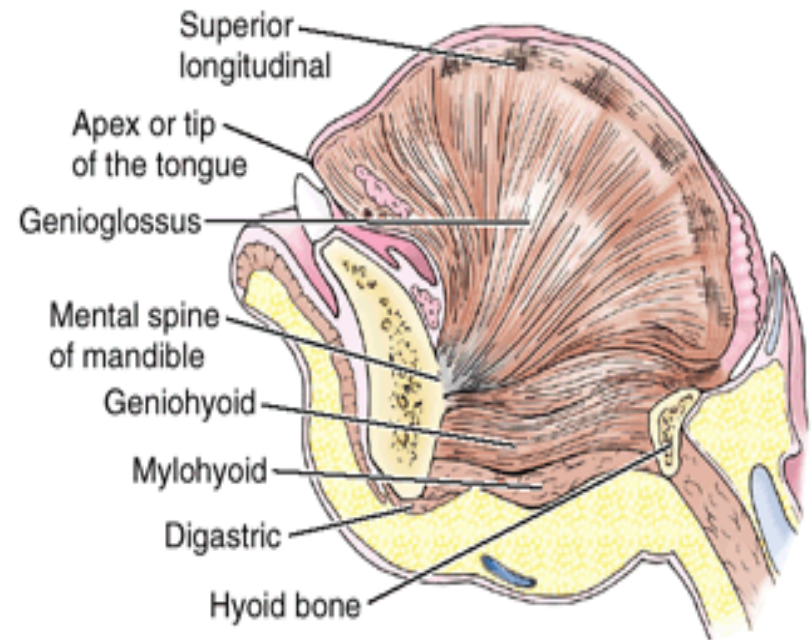
Septum

**Intrinsic
muscles**

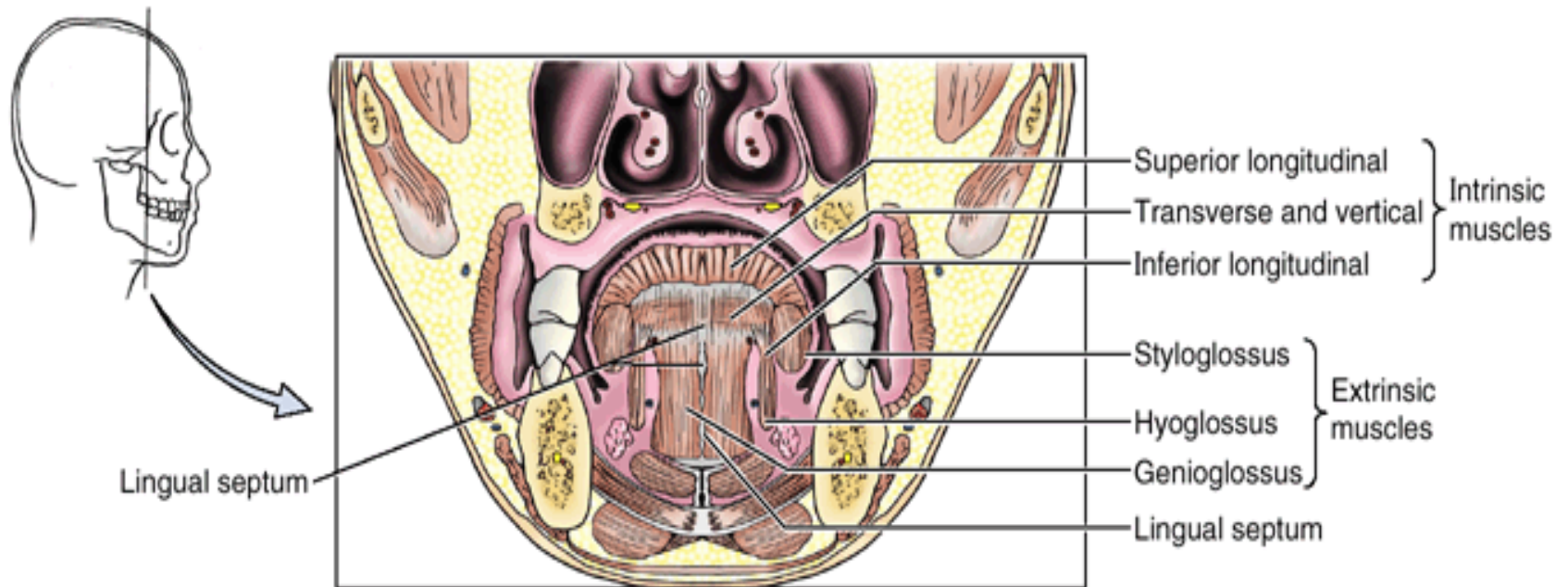




(A) Lateral view

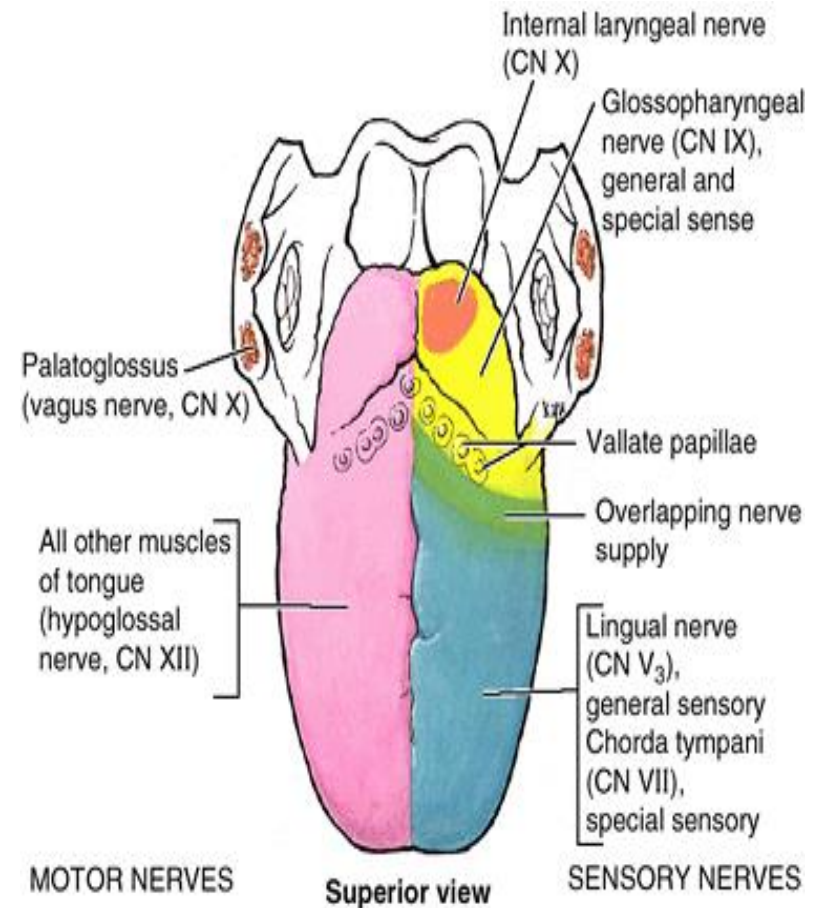
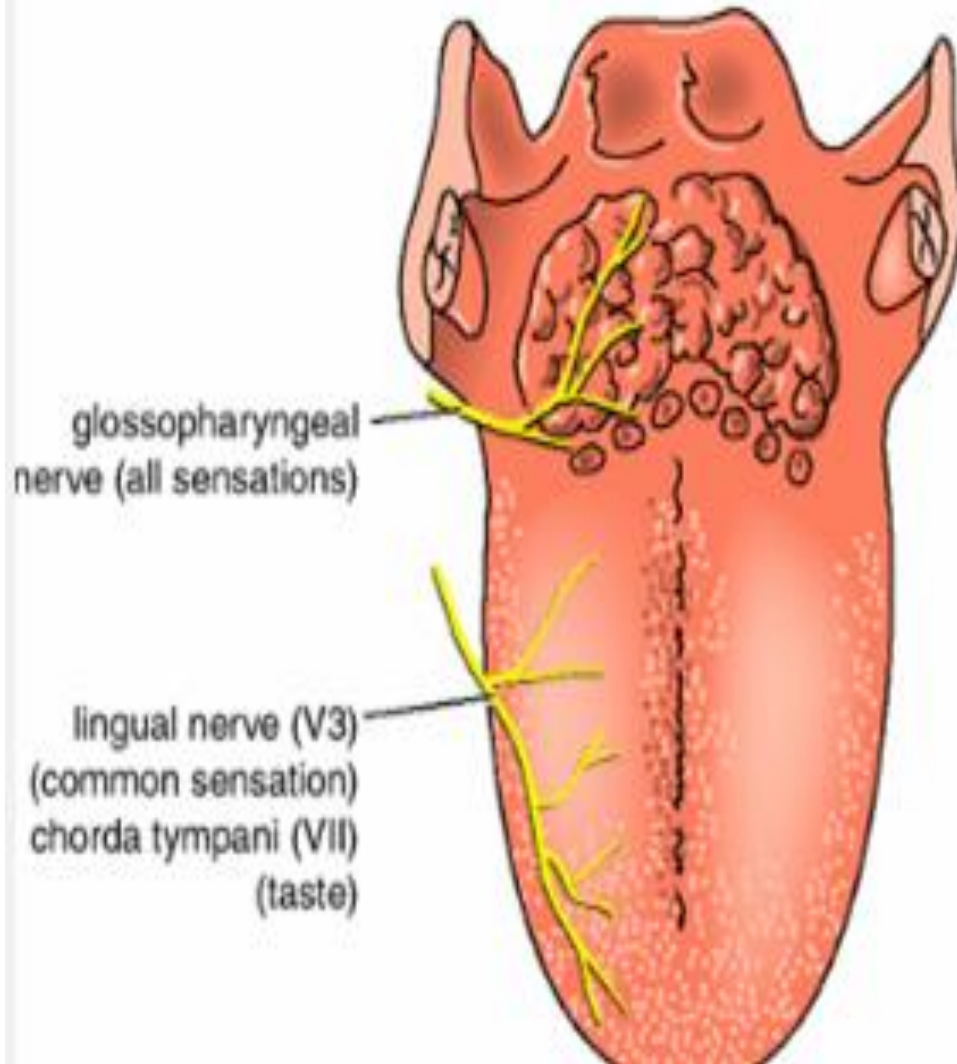


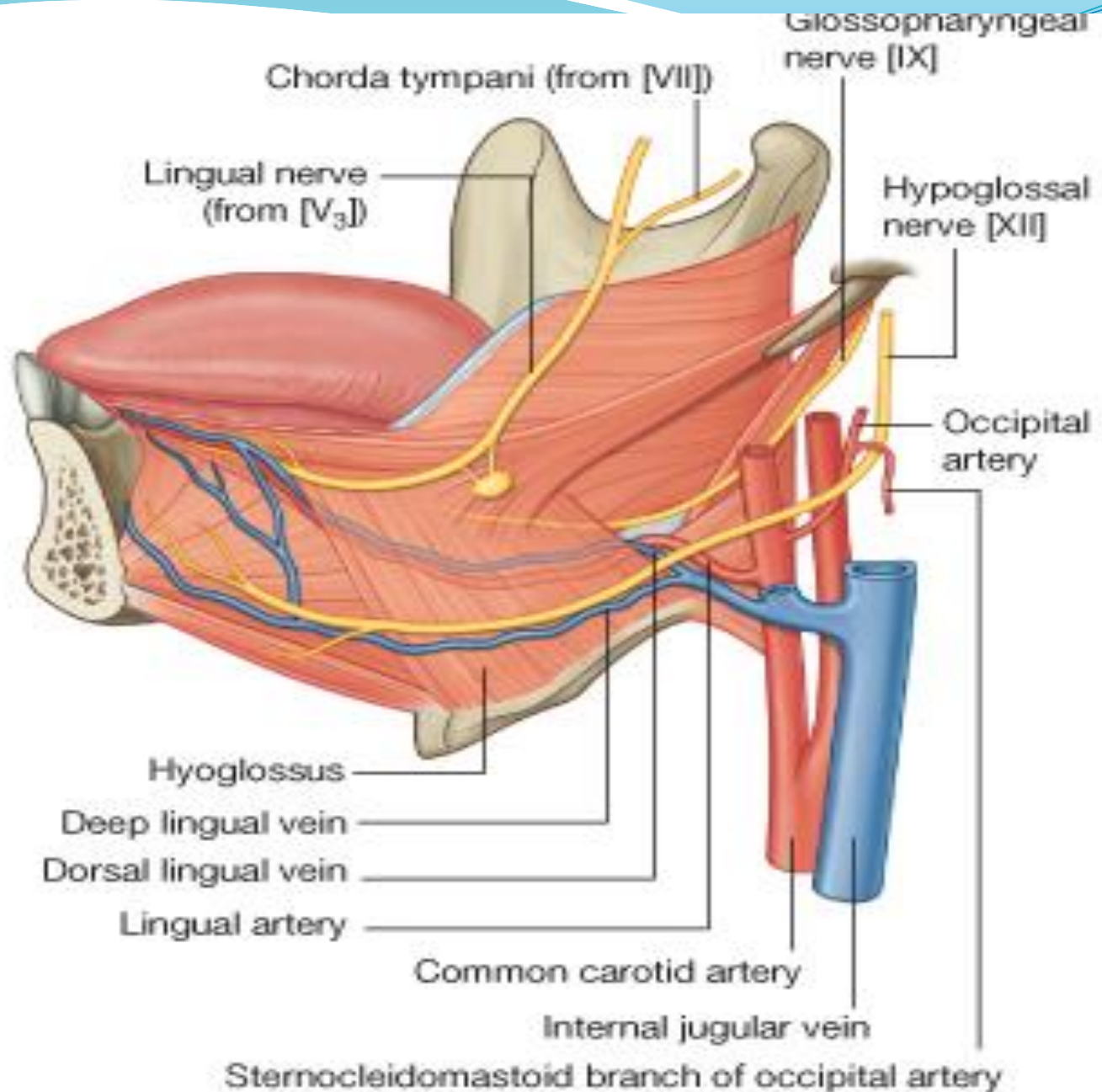
(B) Medial view of right half of bisected tongue



(C) Anterior view of coronal section of mouth

Innervations of the tongue





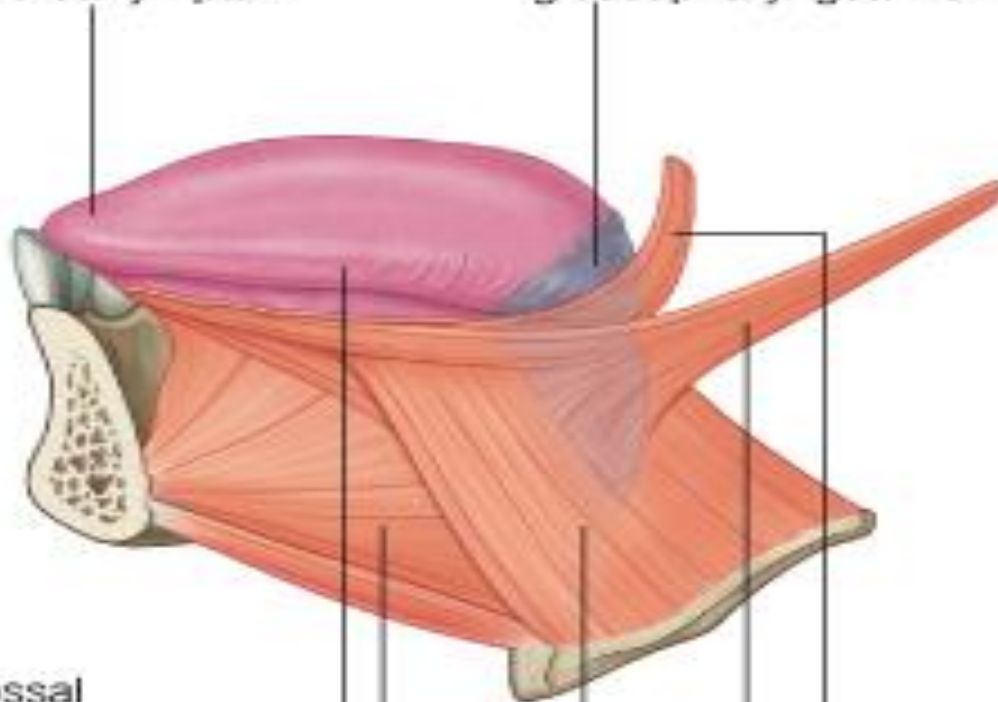
Sensory

Anterior two-thirds (oral)

- general sensation
mandibular nerve [V₃]
via lingual nerve
- special sensation (taste)
facial nerve [VII] via
chorda tympani

Posterior one-third (pharyngeal)

- general and special
(taste) sensation
glossopharyngeal nerve [IX]



Motor

Hypoglossal
nerve [XII]

Intrinsic muscle

Genioglossus

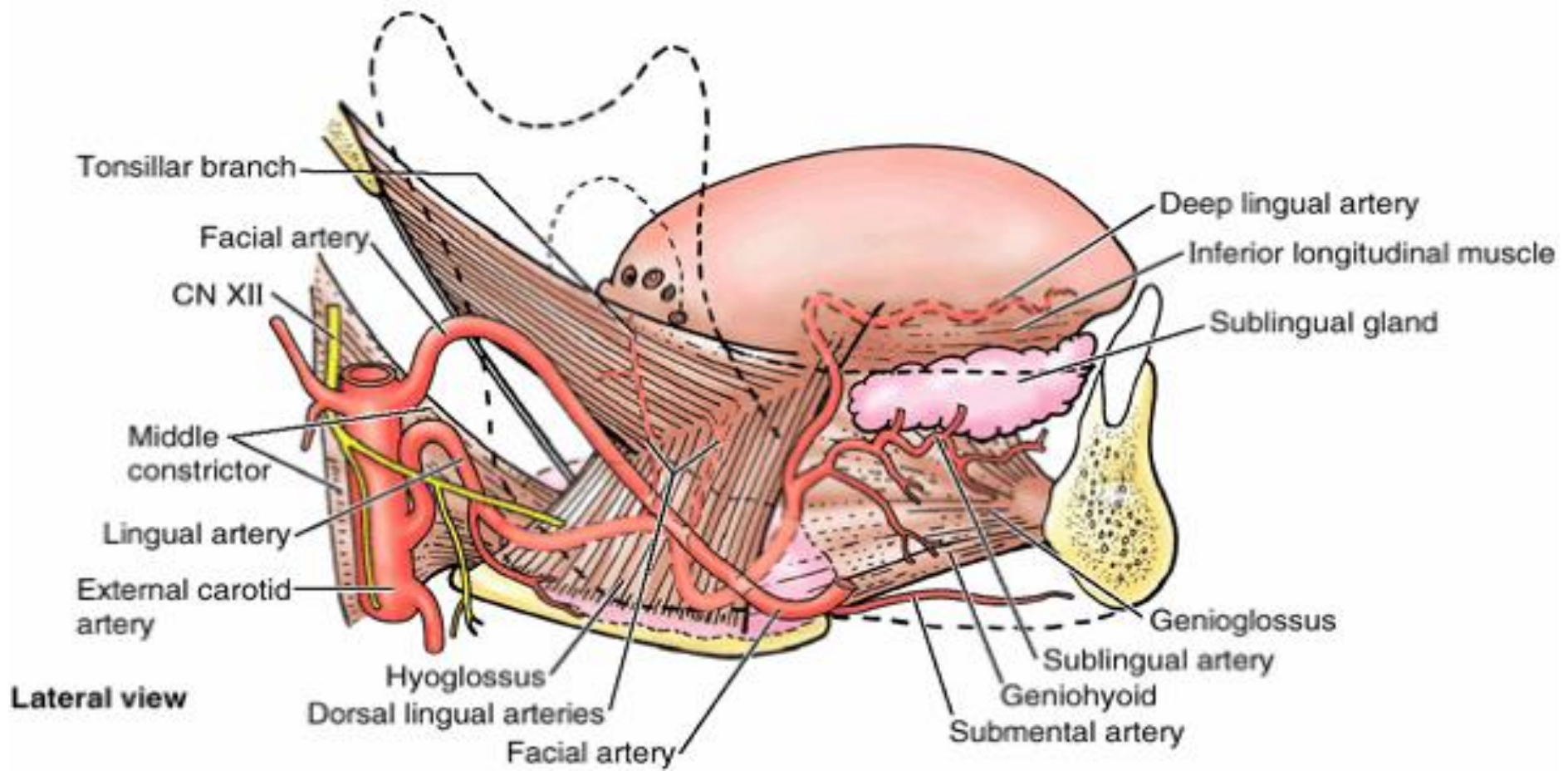
Hyoglossus

Styloglossus

Palatoglossus – vagus nerve [X]

Blood Supply

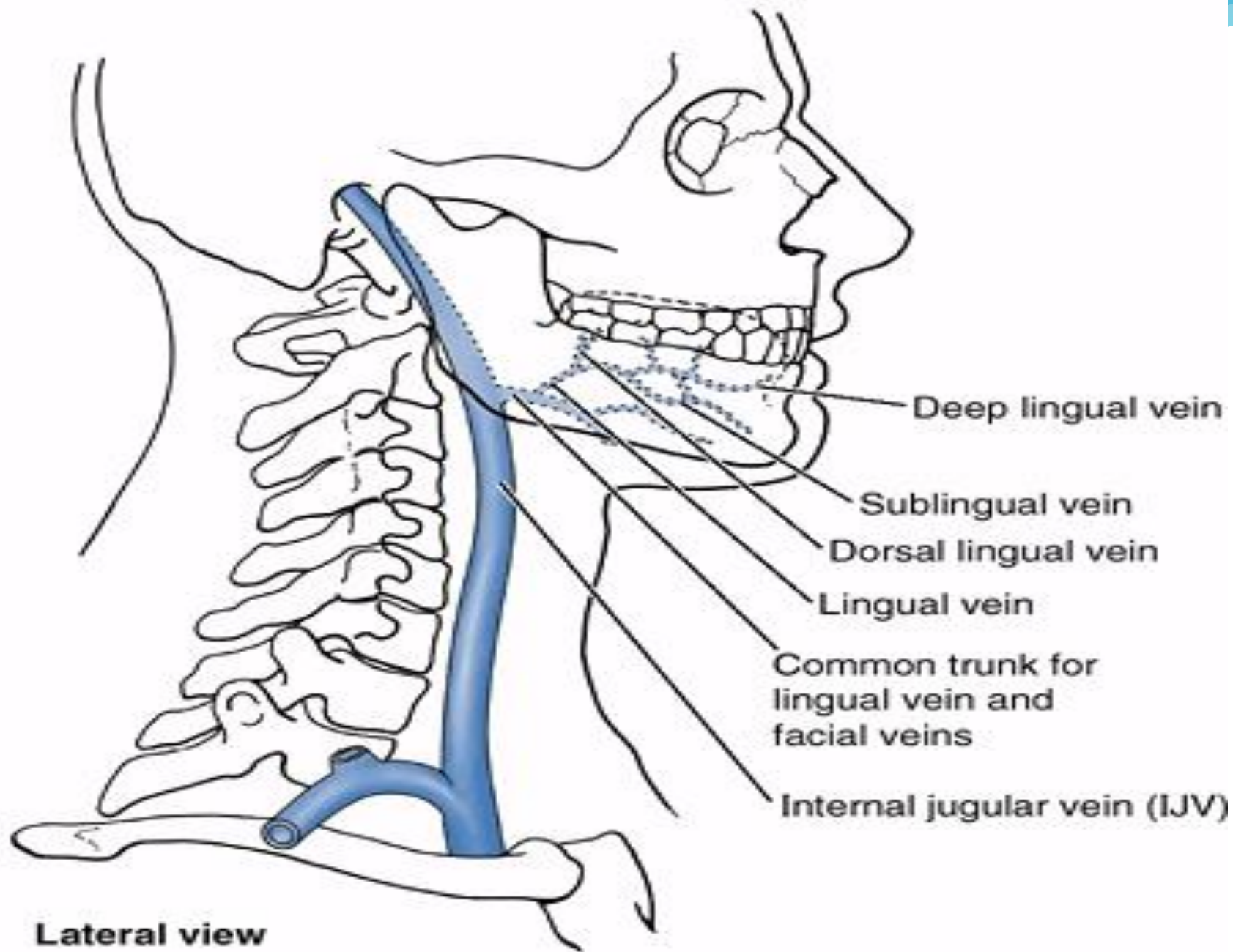
- The lingual artery
- The tonsillar branch of the facial artery
- the ascending pharyngeal artery
- The veins drain into the internal jugular vein



- **The arteries of the tongue:**
 - are derived from the lingual artery, which arises from the external carotid artery
 - On entering the tongue, the lingual artery passes deep to the hyoglossus muscle
 - **The dorsal lingual arteries** supply the posterior part (root)
 - **the deep lingual arteries** supply the anterior part
 - The deep lingual arteries communicate with each other near the apex of the tongue.
 - The dorsal lingual arteries are prevented from communicating by the lingual septum

• The veins of the tongue:

- are the dorsal lingual veins, which accompany the lingual artery
- the deep lingual veins, which begin at the apex of the tongue, run posteriorly beside the lingual frenulum to join the sublingual vein
- The sublingual veins in elderly people are often varicose (enlarged and tortuous).
- All these lingual veins terminate, directly or indirectly, in the IJV.



Lateral view

Lymph Drainage

- is exceptional.
- Most of the lymphatic drainage converges toward and follows the venous drainage
- lymph from the tip of the tongue, frenulum, and central lower lip runs an independent course
- Lymph from the tongue takes four routes
 - Lymph from the posterior third
 - drains into the superior deep cervical lymph nodes.
 - Lymph from the medial part of the anterior two thirds
 - drains directly to the inferior deep cervical lymph nodes.

- Lymph from the lateral parts of the anterior two thirds
 - drains to the submandibular lymph nodes.
- The apex and frenulum:
 - drain to the submental lymph nodes.
- The posterior third and the medial part of the anterior two thirds drain bilaterally

RULES OF NERVE SUPPLY FOR MUSCLE GROUPS

ALL MUSCLES OF	SUPPLIED BY	EXCEPT	WHICH IS SUPPLIED BY
PHARYNX	Pharyngeal plexus (IX, X & sympathetic)	Stylopharyngeus	Glossopharyngeal (IX)
PALATE	Pharyngeal plexus (IX, X & sympathetic)	Tensor veli palatini	Nerve to medial pterygoid (Vc)
TONGUE	Hypoglossal (XII)	Palatoglossus	Pharyngeal plexus (IX, X & sympathetic)
FACIAL EXPRESSION & BUCCINATOR	Facial (VII)	Levator palpebrae superioris	Oculomotor (III)
MASTICATION	Mandibular division of Trigeminal (Vc)	Buccinator	Facial (VII)
LARYNX	Recurrent laryngeal	Cricothyroid	External branch of superior laryngeal nerve (X)



Thanks